

Faculté des sciences et technologies

TROISIEME ANNEE

Rapport du travail de laboratoire no 3

Cours : Reseau1

Étudiant: Jouvens JOANEL

Enseignant: Ismaël SAINT AMOUR

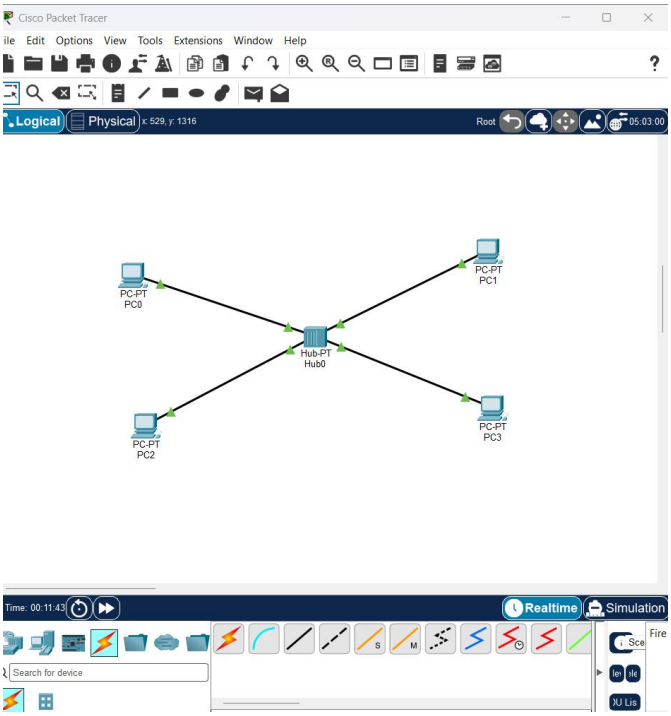
Date: 02 novembre 2026

L’objectif de ce TD est de :

1. Savoir attribuer des adresses IP valides aux machines.
2. Comprendre l’adressage IPv4 et IPv6.
3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.
4. Vérifier la connectivité avec les commandes ping et ping ipv6.

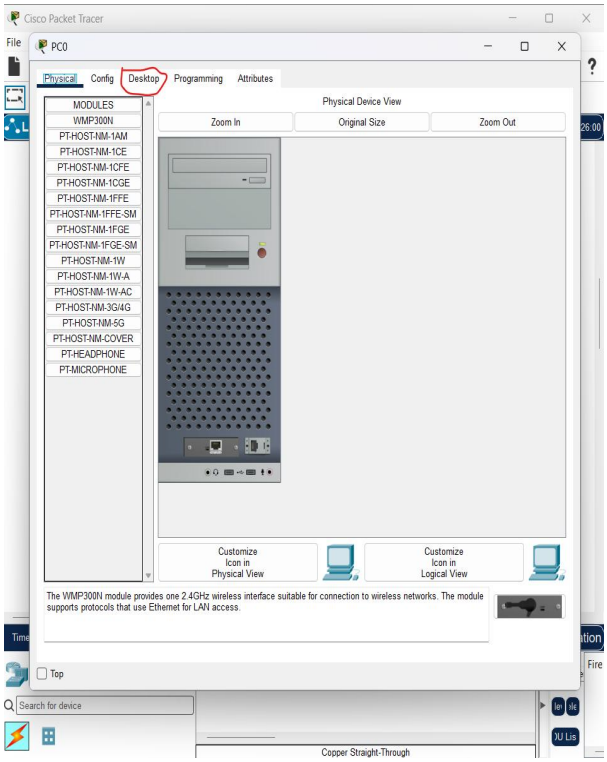
Etapes du TD

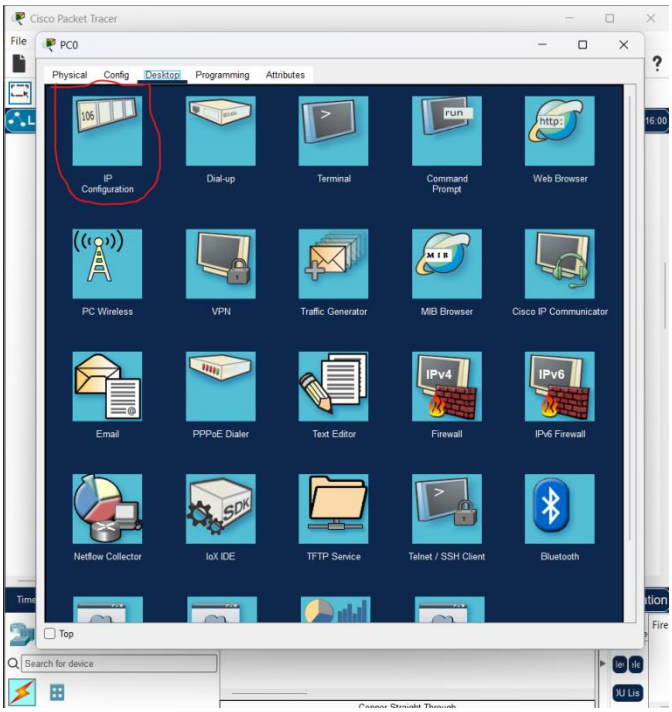
1) Création d'un topologies avec 4 PC puis un Hub



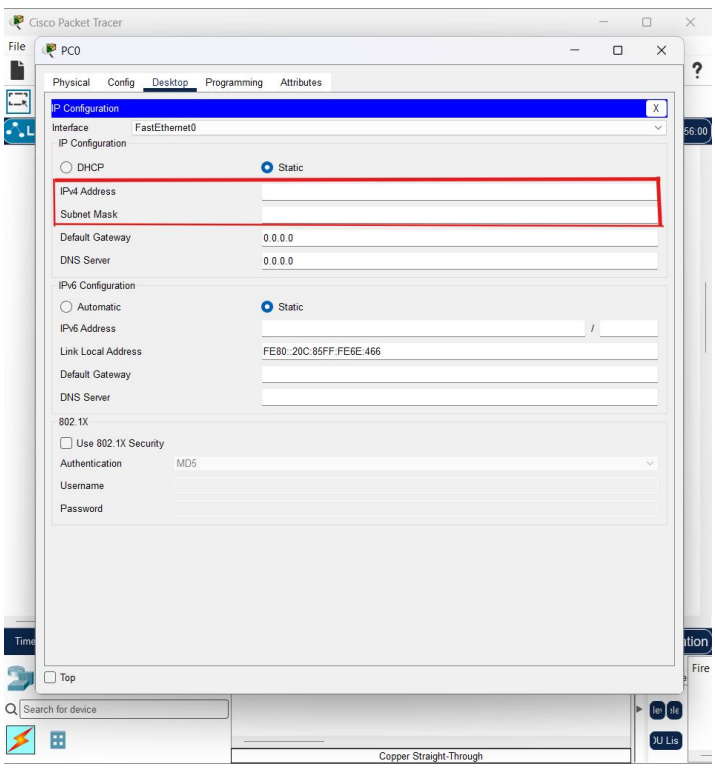
2) configuration en ip4

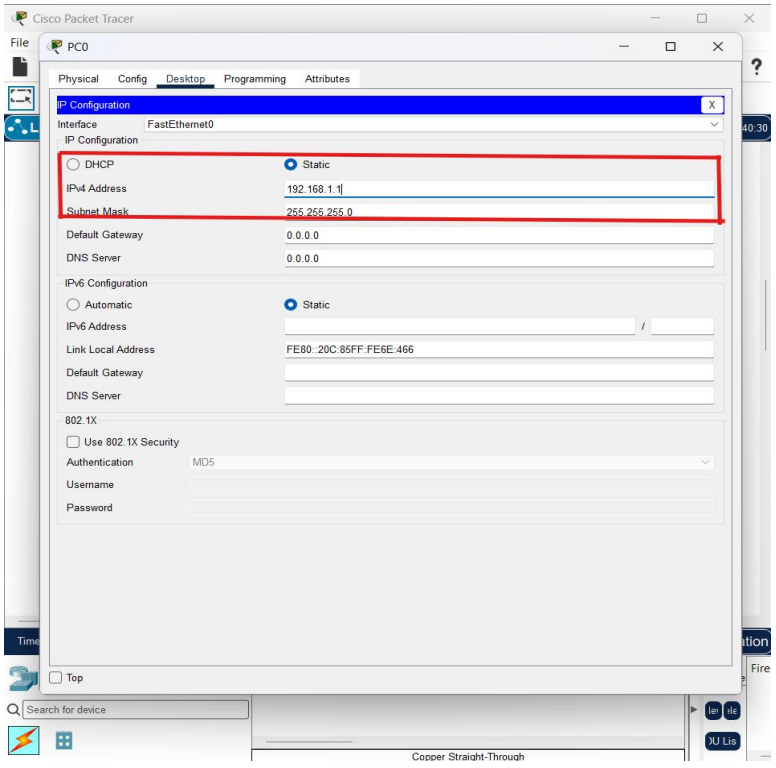
a) interface du paramètre des PC

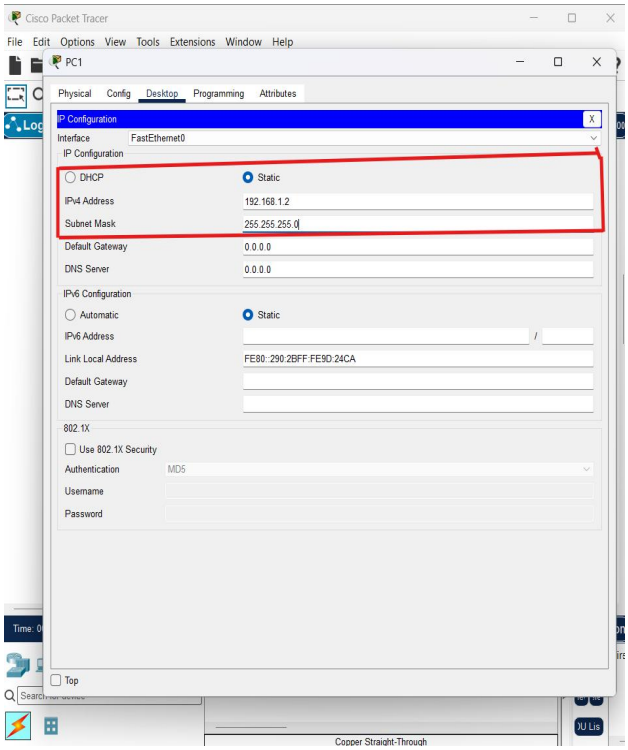


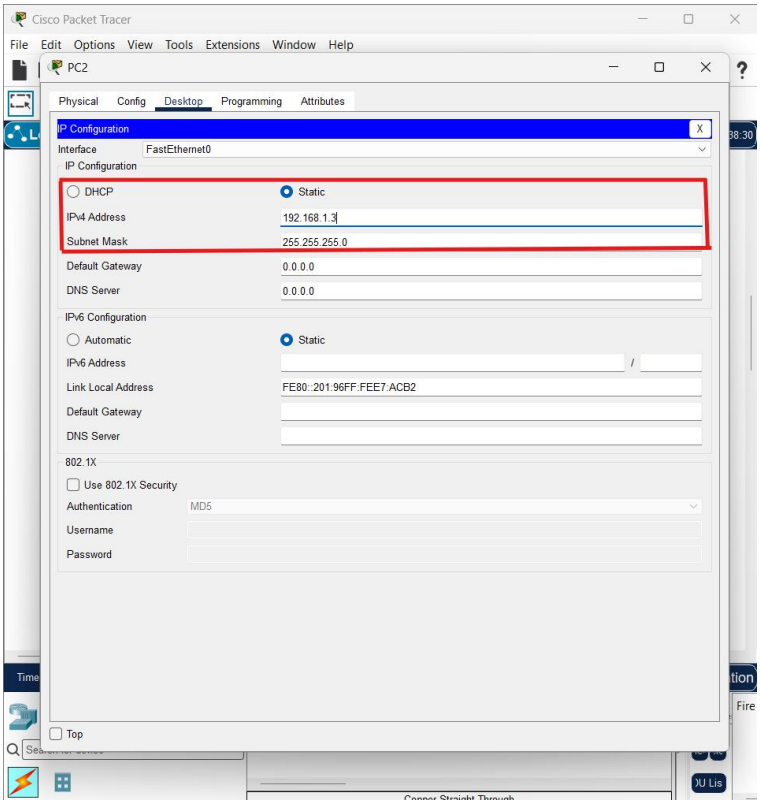


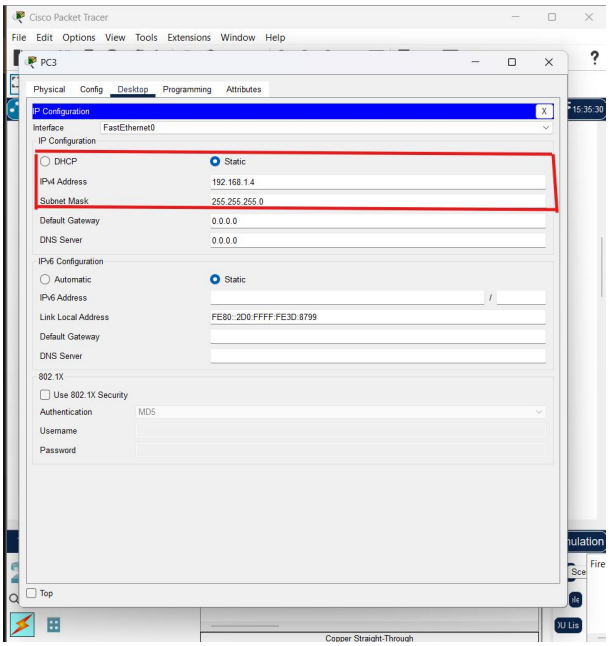
b) association des adresses ipv4 dans les PC



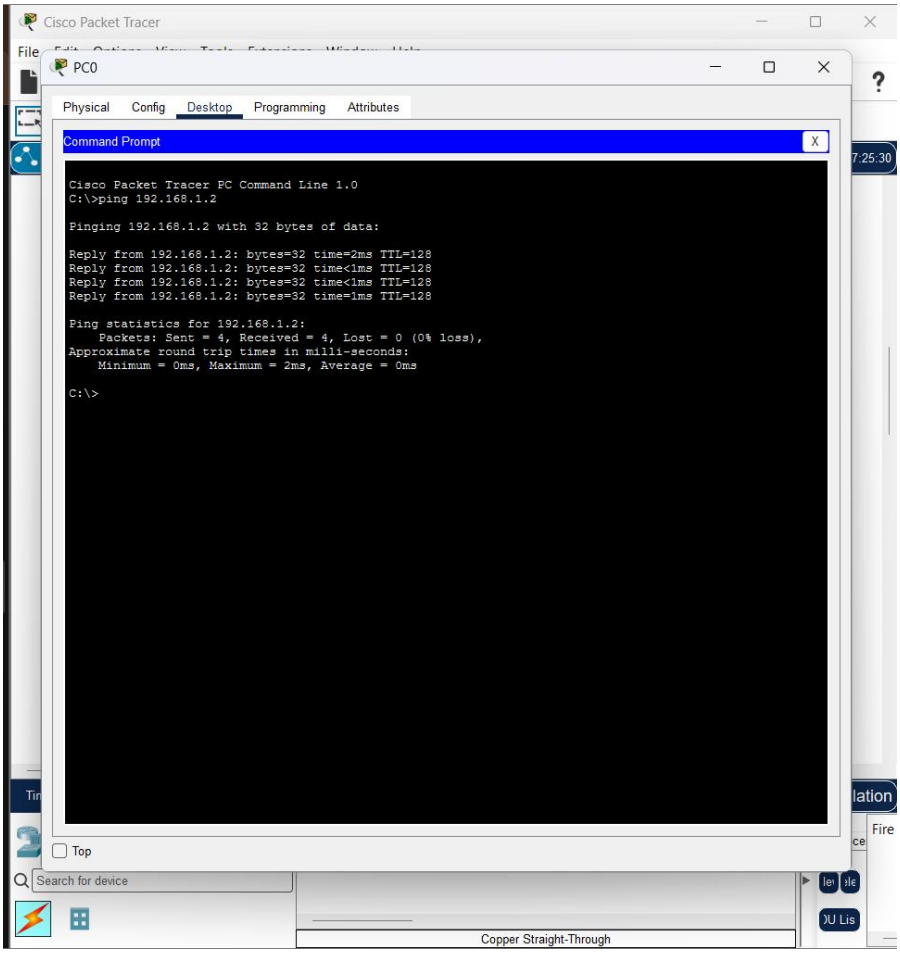




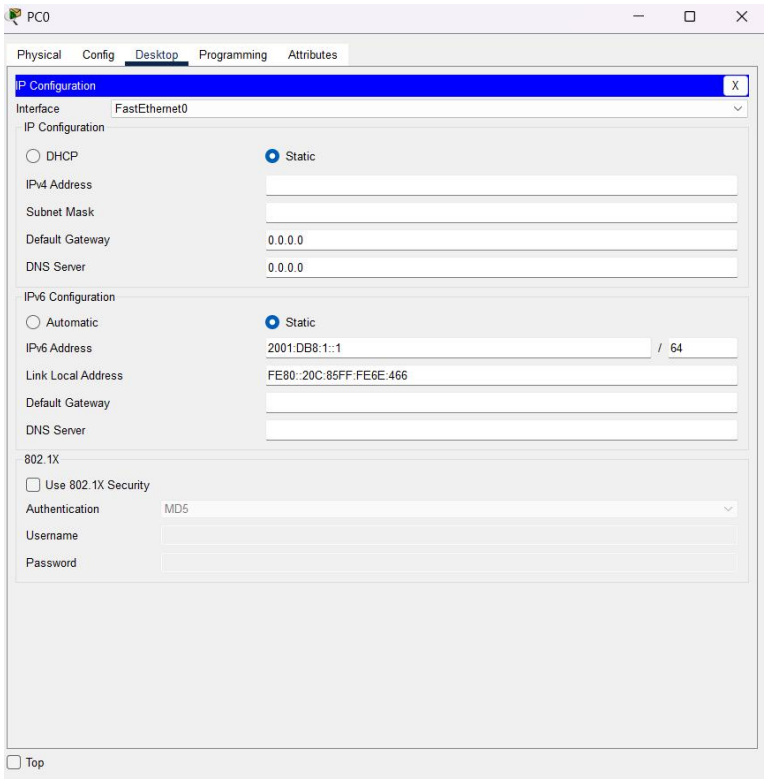


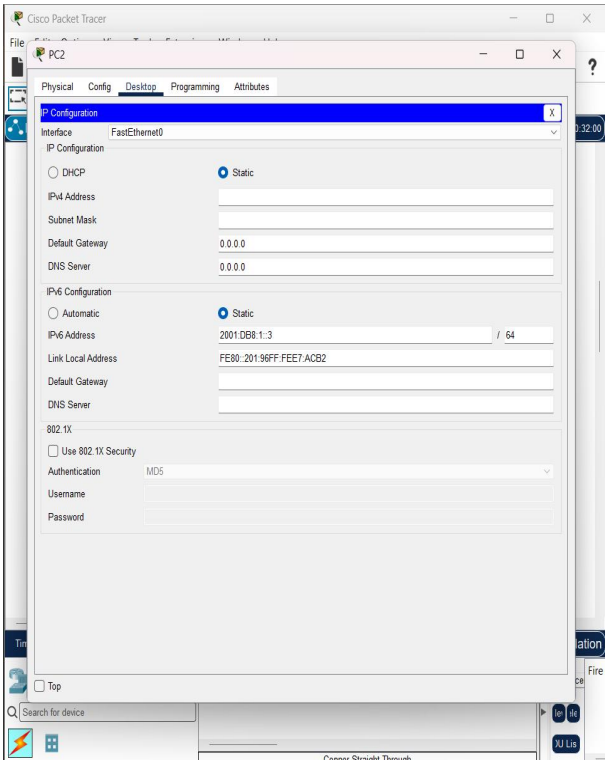


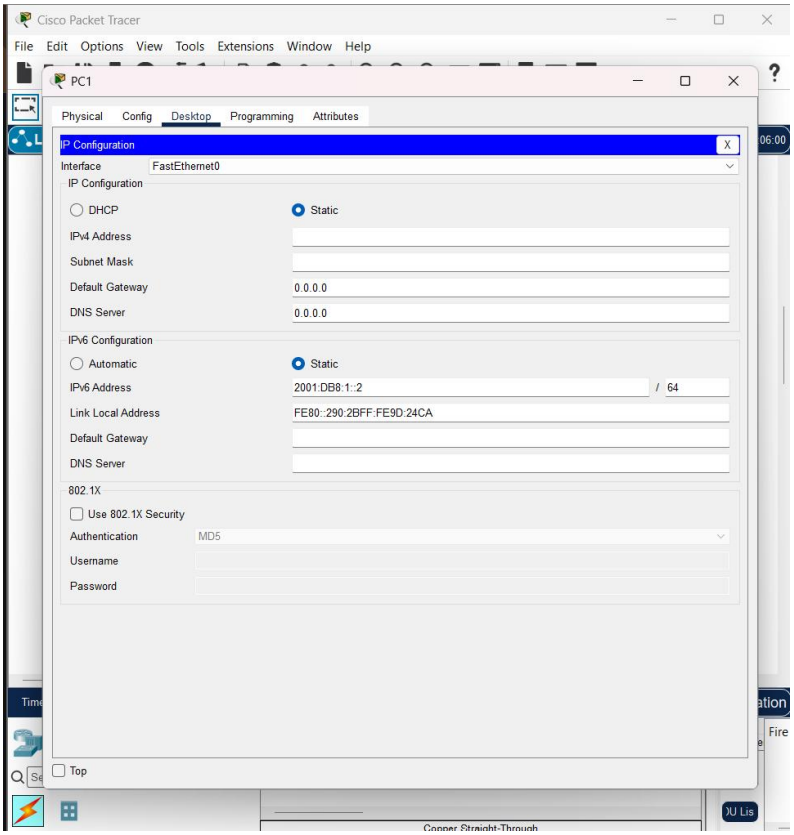
3) vérification de la connexion entre les PC avec le command ping plus l'adresse ipv4

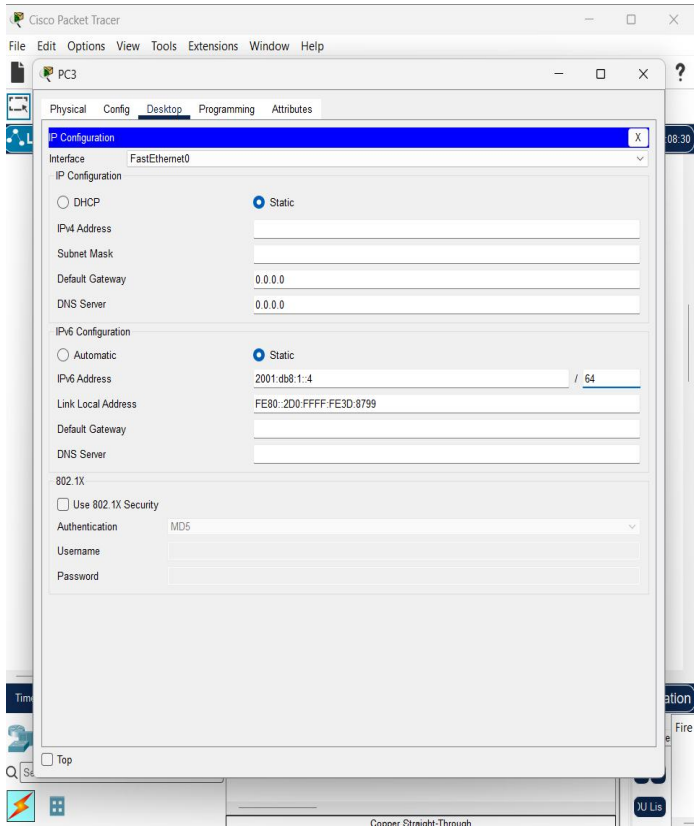


4) Configuration de l'ipv6

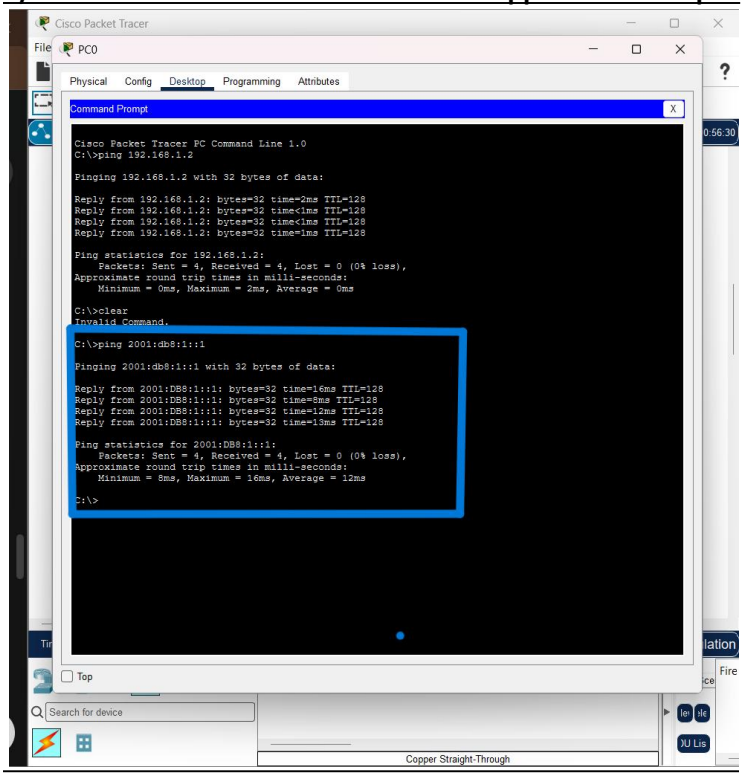

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::1
/ 64
Link Local Address
FE80::20C:85FF:FE6E:466
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Cisco Packet Tracer
PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::3
/ 64
Link Local Address
FE80::201:96FF:FEE7:ACB2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search for device


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::2
/ 64
Link Local Address
FE80::290:2BFF:FE9D:24CA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

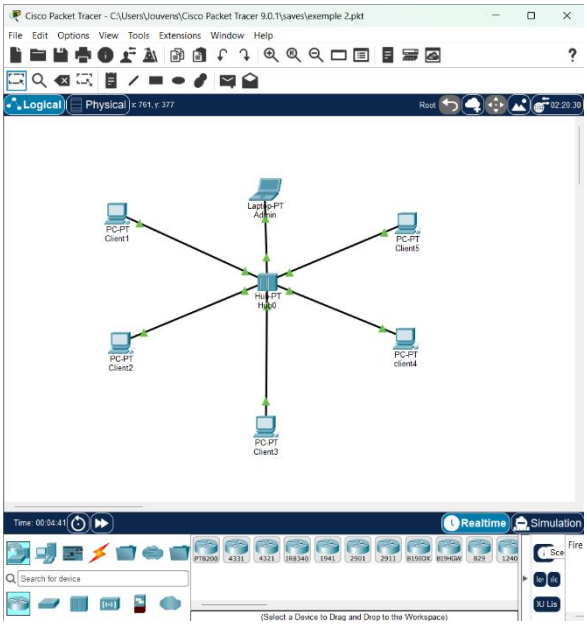

Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:db8:1::4
/ 64
Link Local Address
FE80::2D0:FFFF:FE3D:8799
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

5) Vérification de la connexion entre les appareils avec l'ipv6

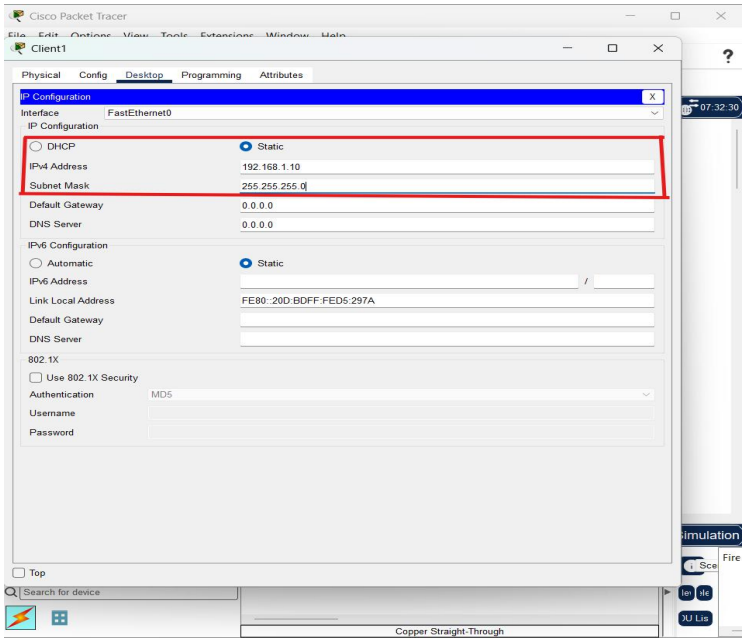


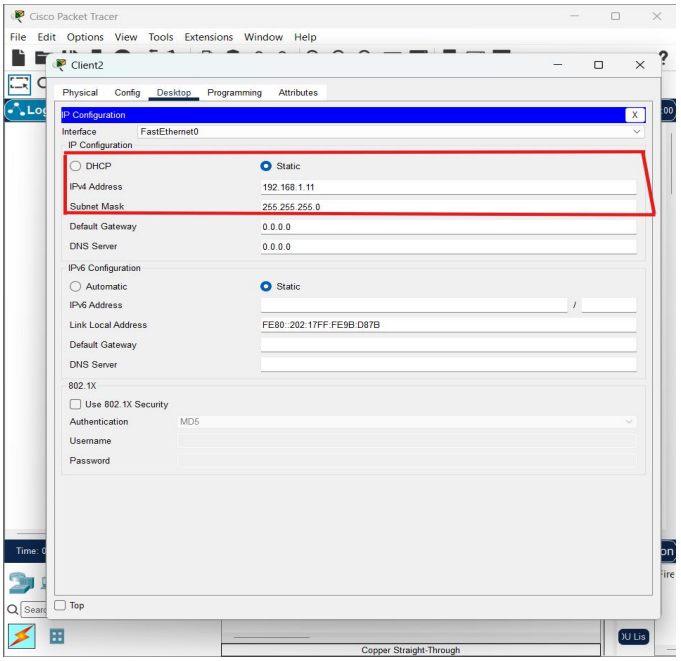
Travaux dirigées

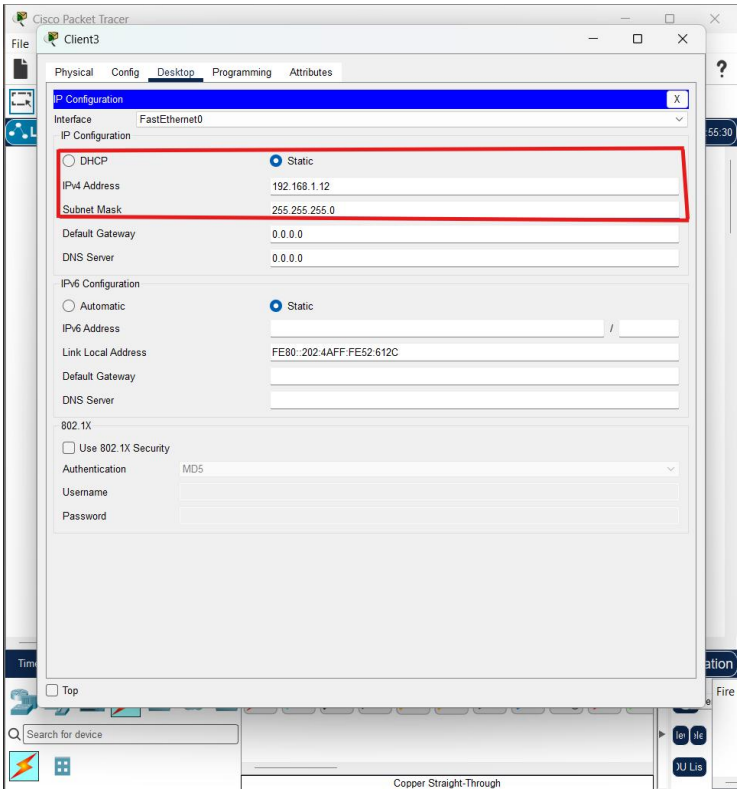
1- Reproduction de la topologie puis la configuration de l'adresses IPv4

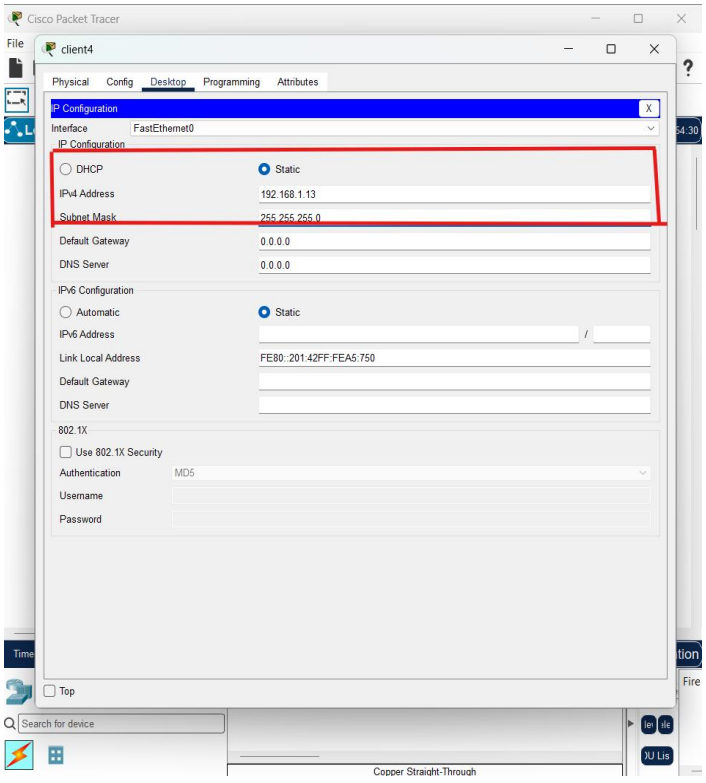


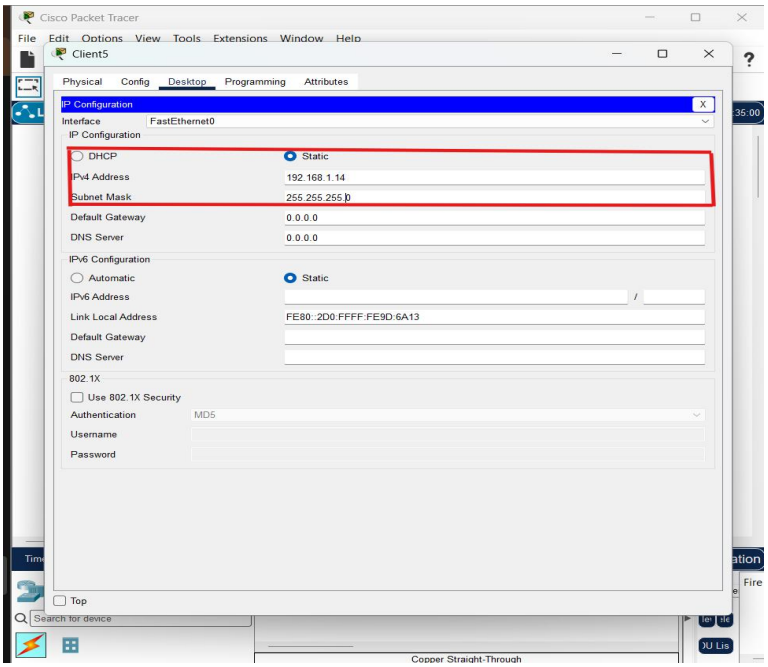
a) Association de l'adresses IPv4 a des PC

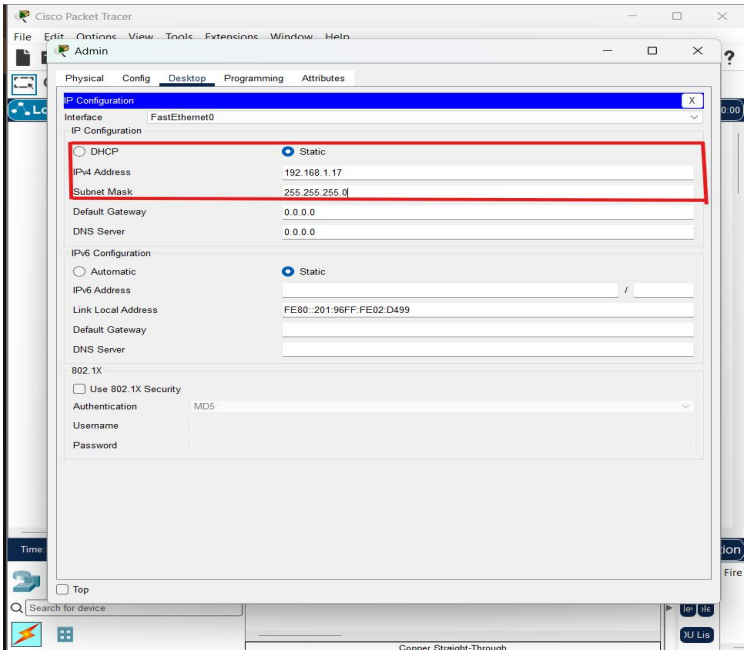

Cisco Packet Tracer
Client1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::20D:BDFF:FED5:297A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search for device
Copper Straight-Through


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Client2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:17FF:FE9B:D87B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Copper Straight-Through

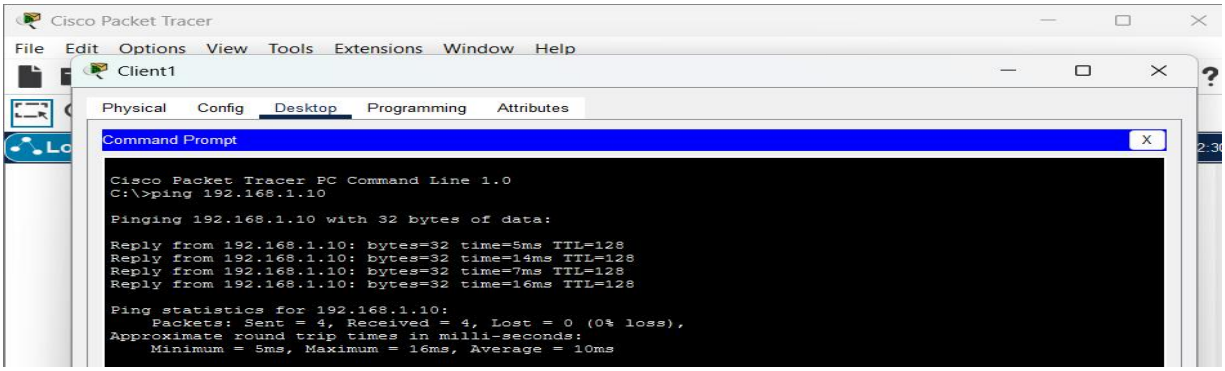

Cisco Packet Tracer
Client3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:4AFF:FE52:612C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search for device
Copper Straight-Through


Cisco Packet Tracer
client4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:42FF:FEA5:750
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Search for device
Copper Straight-Through

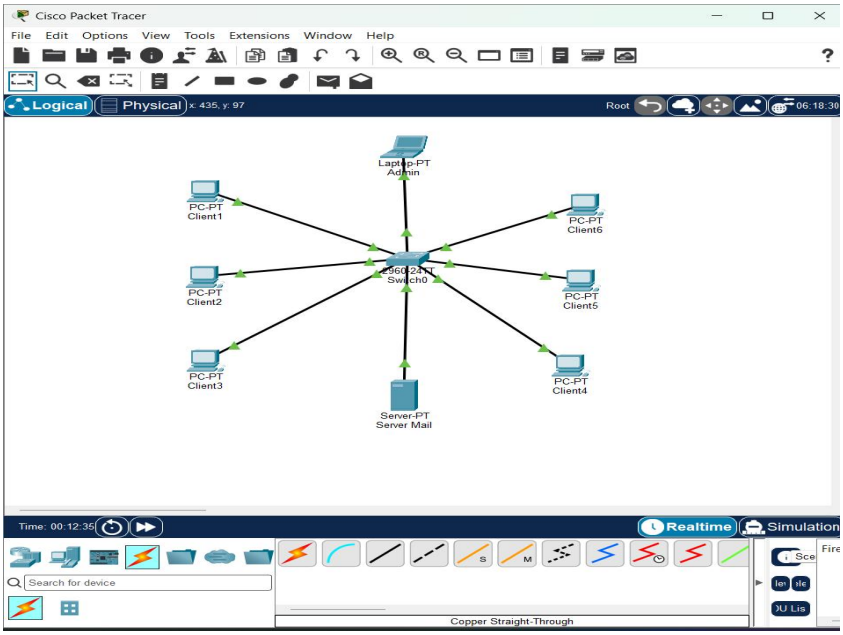




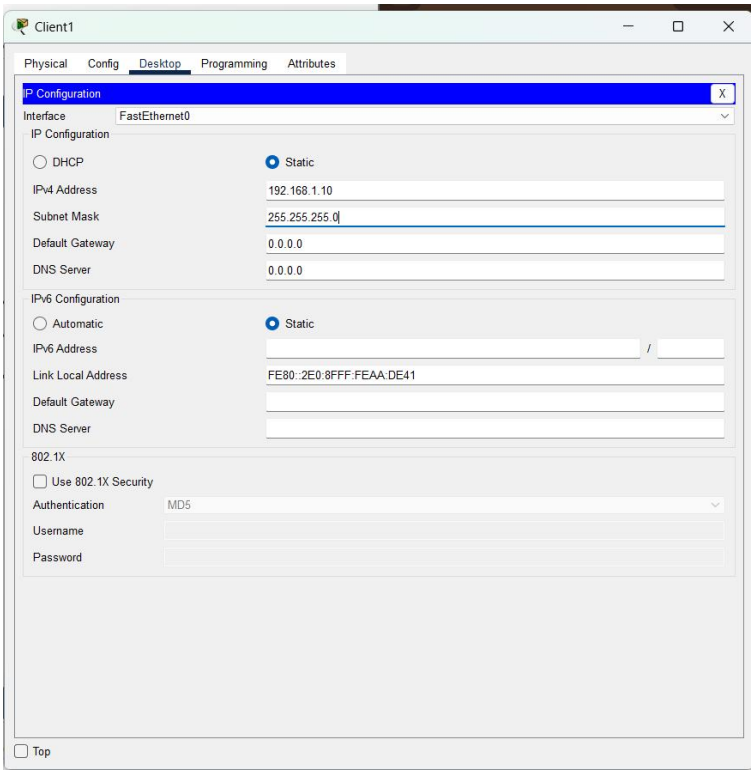
b) Vérification de la connexion entre les appareils avec l'IPv4

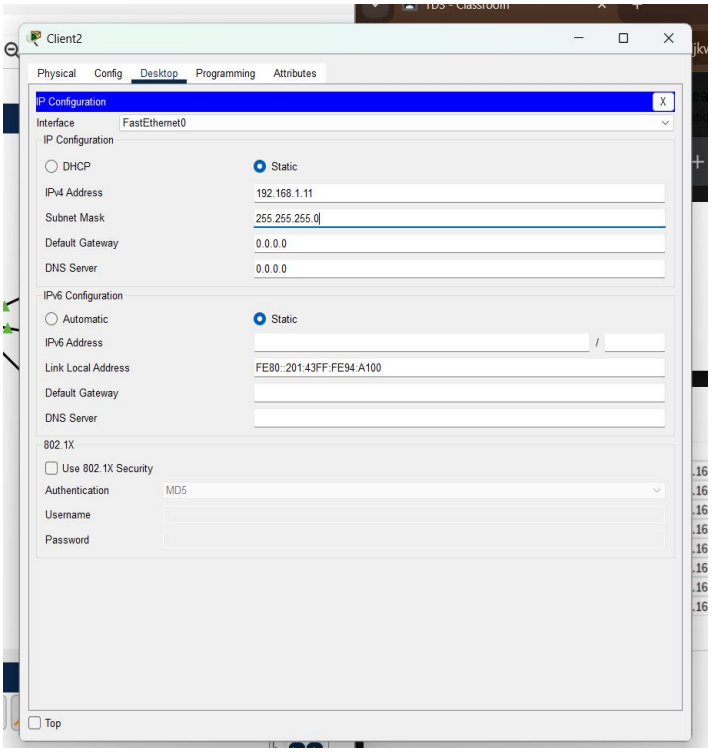


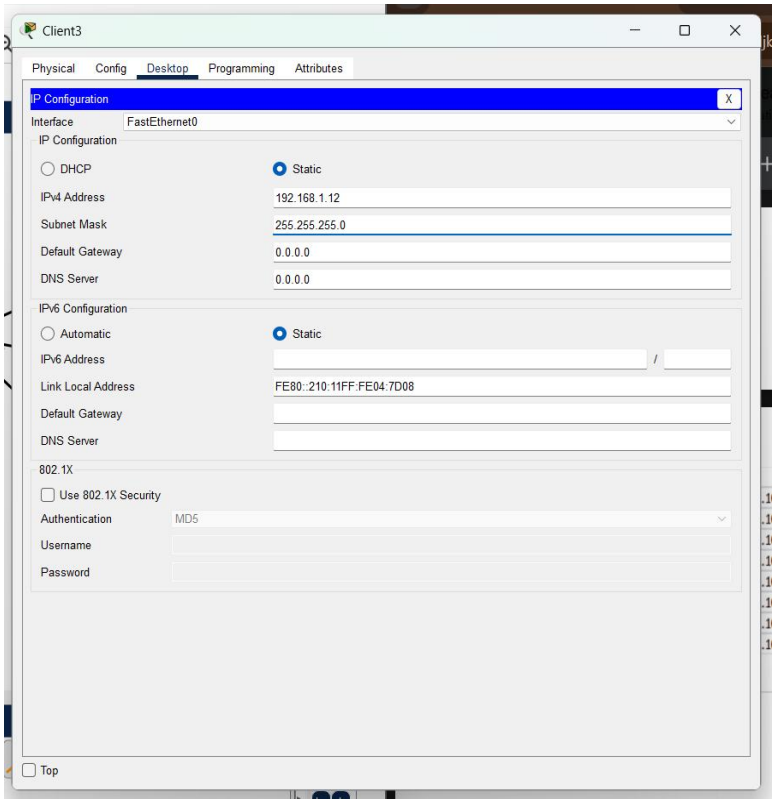
2-Reproduction de la topologie puis configuration de l'adresses IPv4

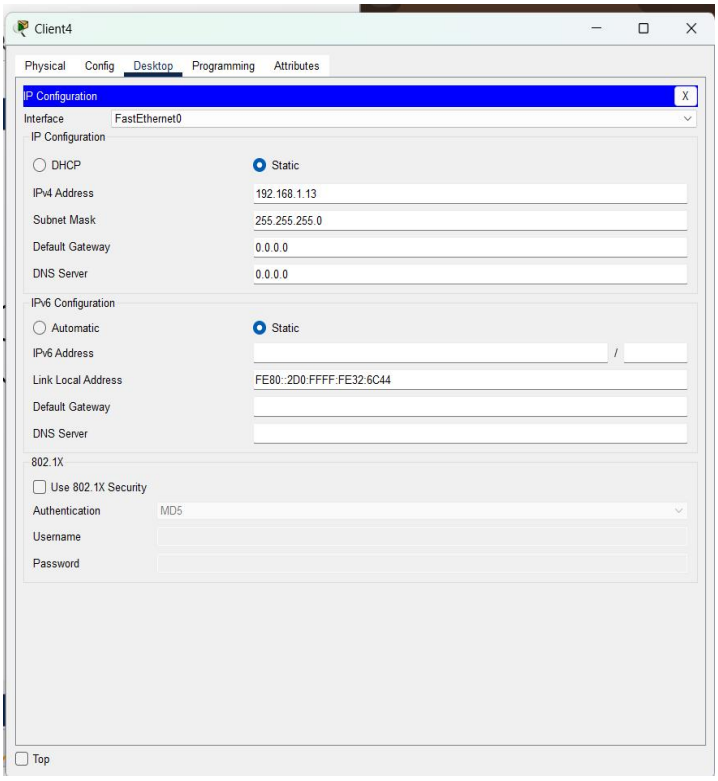


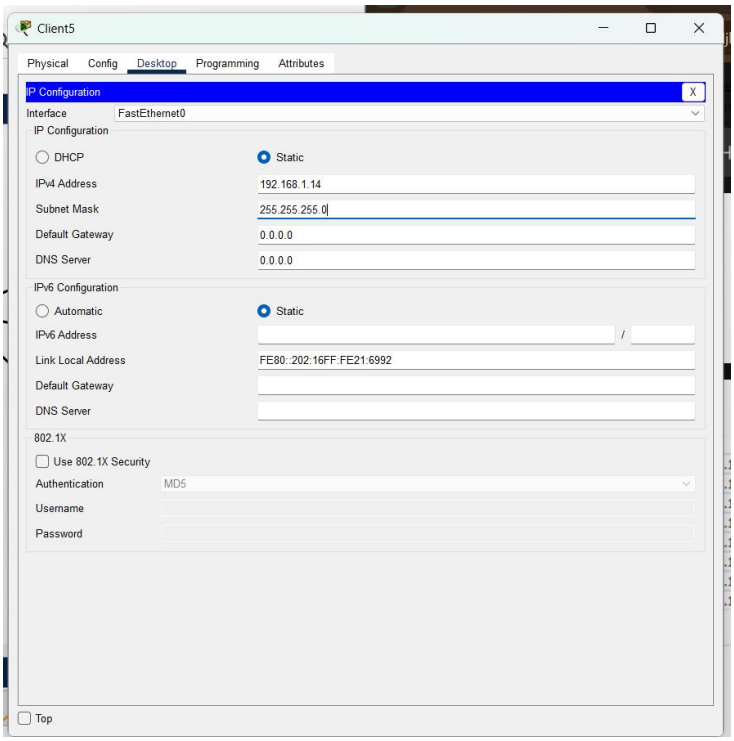
a) Association de l'adresses IPv4

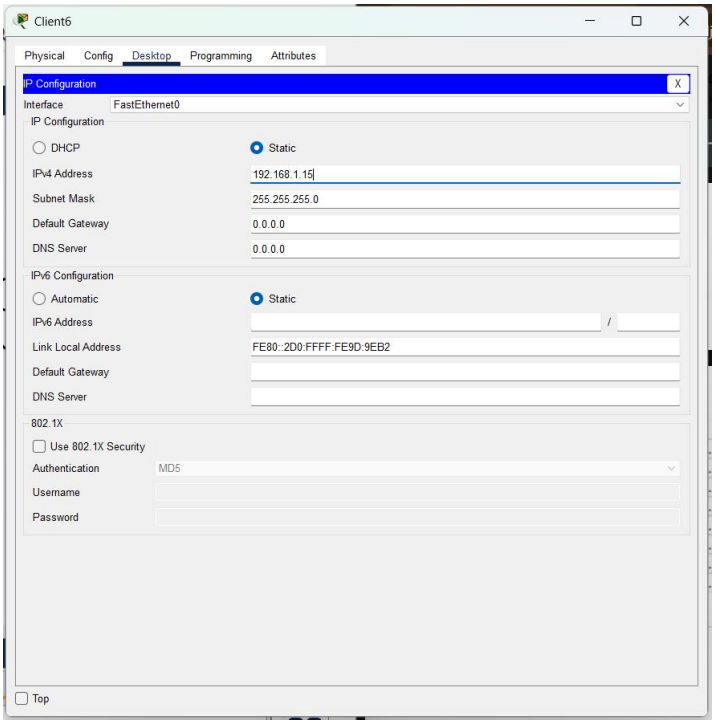


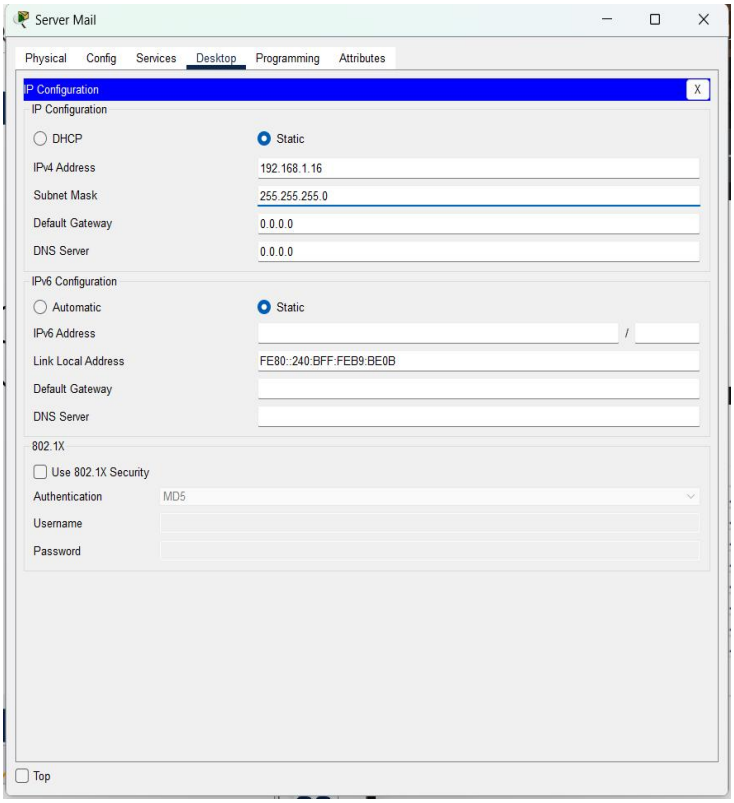


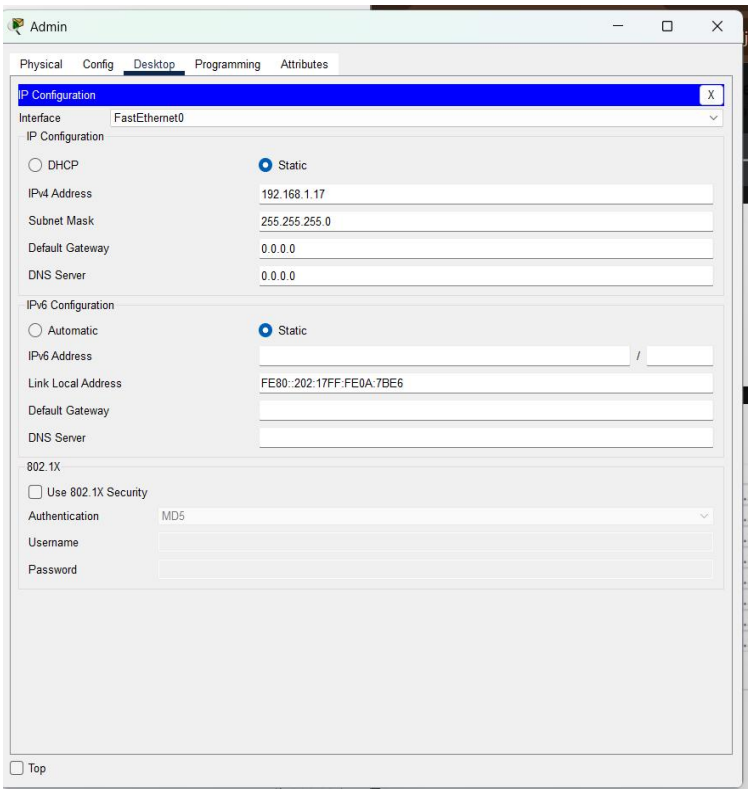





Client5
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.14
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::202:16FF:FE21:6992
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top


Client6
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.15
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::2D0:FFFF:FE9D:9EB2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top


Server Mail
Physical Config Services Desktop Programming Attributes
IP Configuration
IP Configuration
DHCP Static
IPv4 Address 192.168.1.16
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::240:BFF:FEB9:BE0B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top


Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.17
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic Static
IPv6 Address /
Link Local Address FE80::202:17FF:FE0A:7BE6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top

b) Teste de la connexion entre les appareils

```
C:\>ping 3000:159:1::2

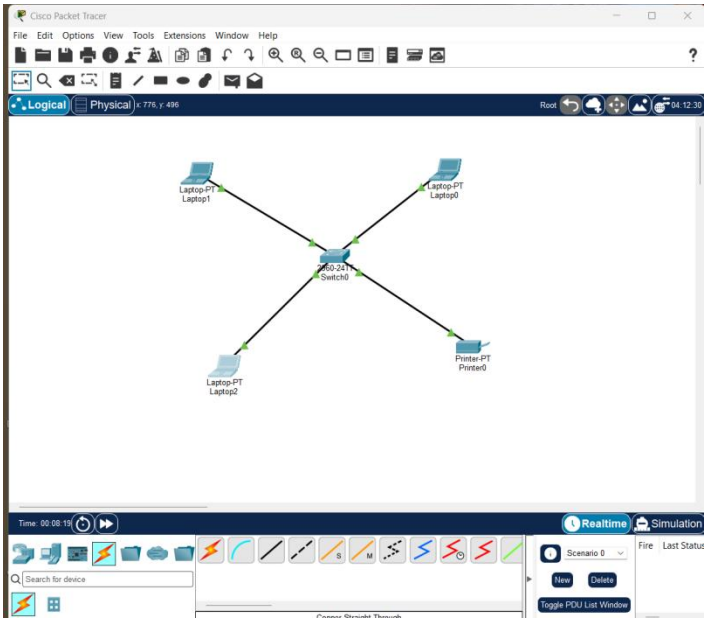
Pinging 3000:159:1::2 with 32 bytes of data:

Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128

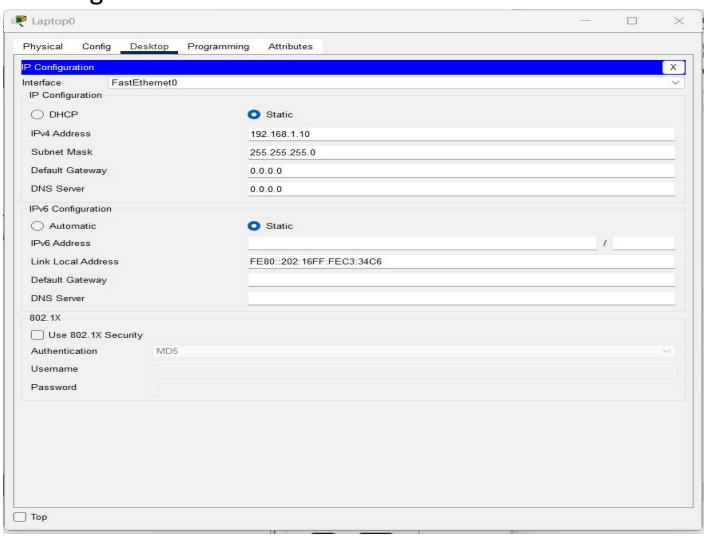
Ping statistics for 3000:159:1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

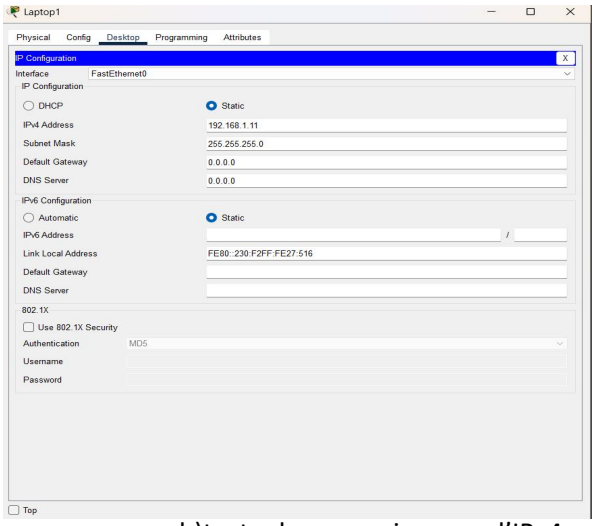
C:\>
```

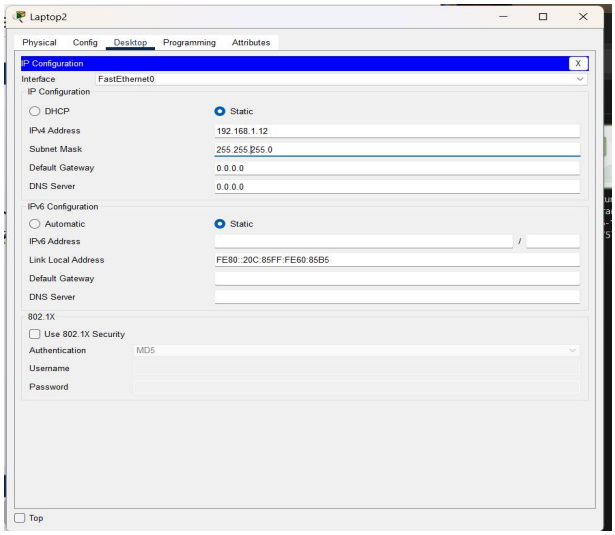
3- parmi les topologies, choisissez-en une, configurant les adresses IPv4 et tester la connectivité



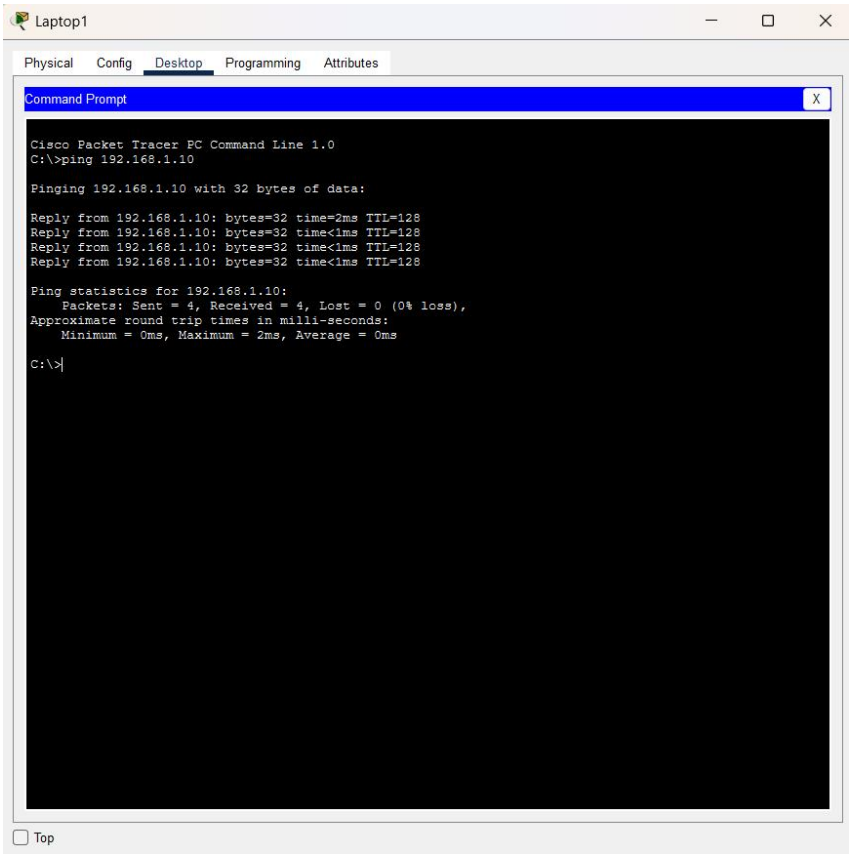
a) la configuration de l'adresses IPv4



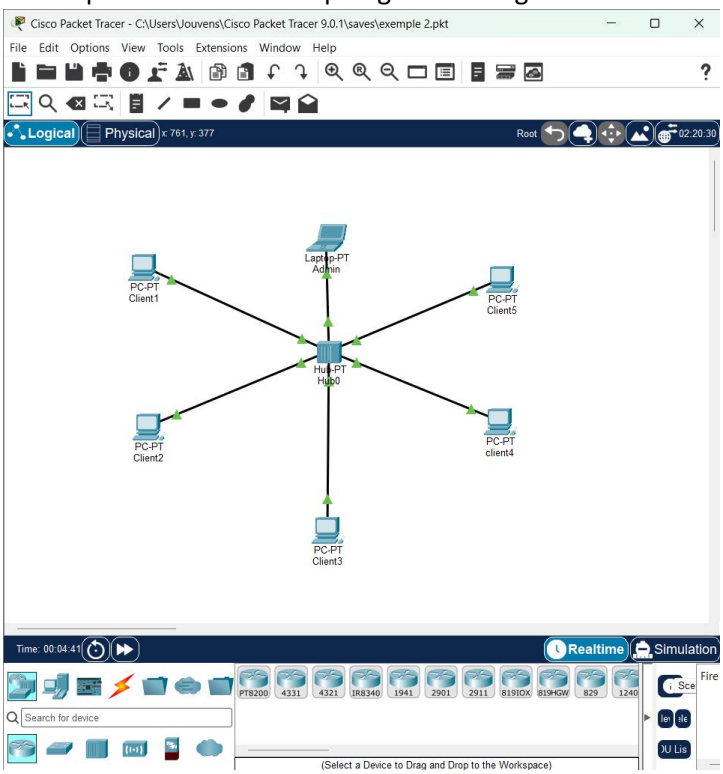




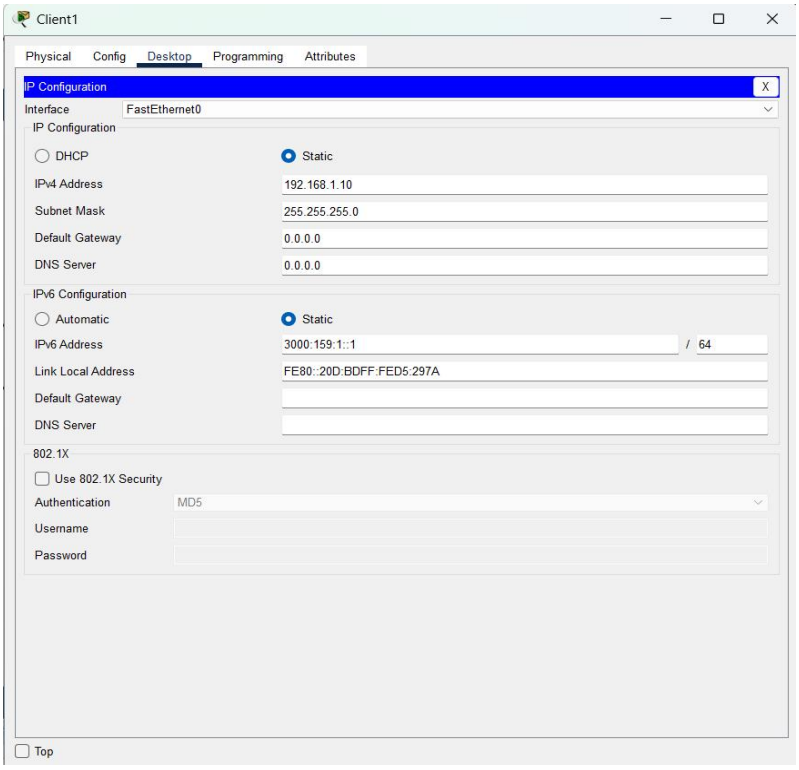
b)teste de connexion avec l'IPv4

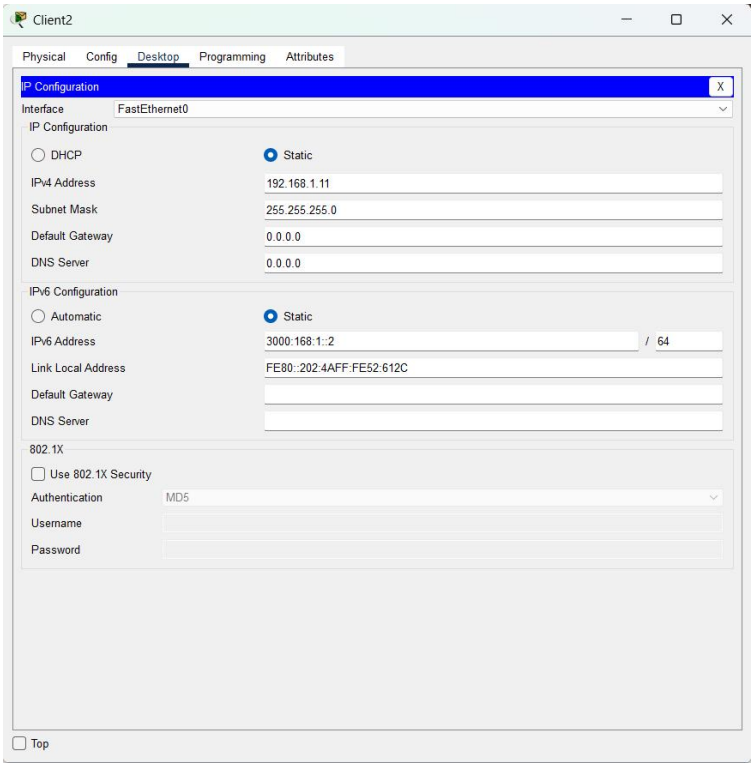


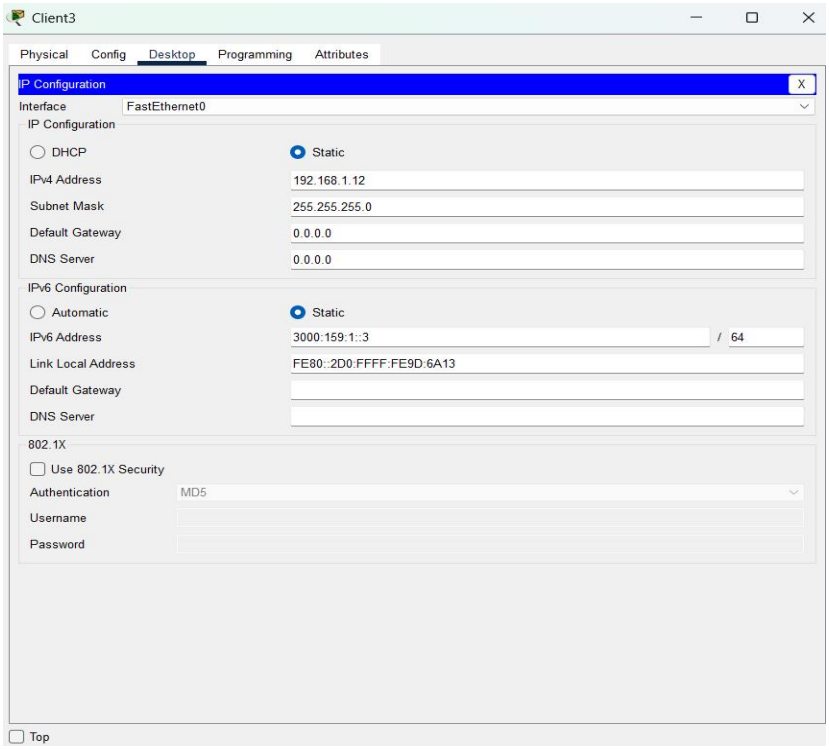
4- la reproduction d’une topologie en configurant les adresses en IPv6

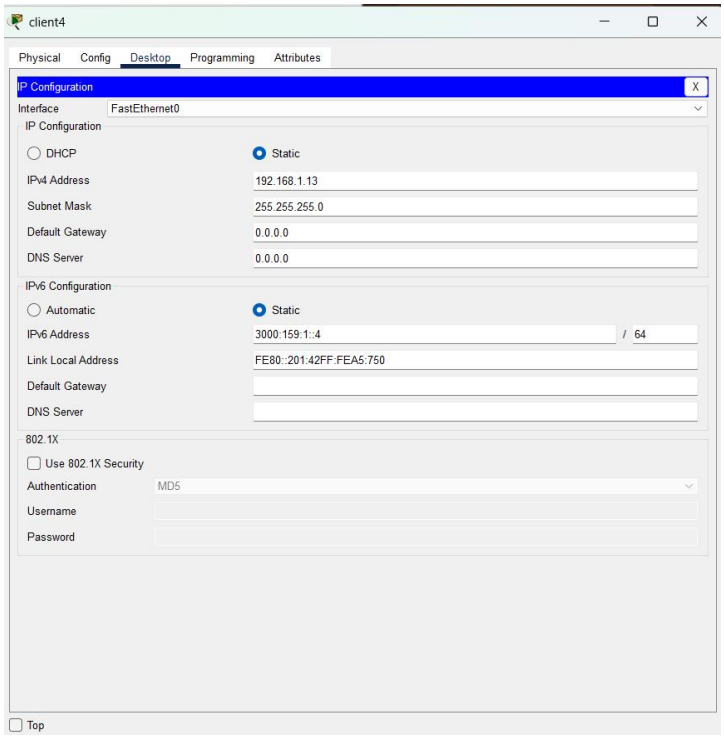


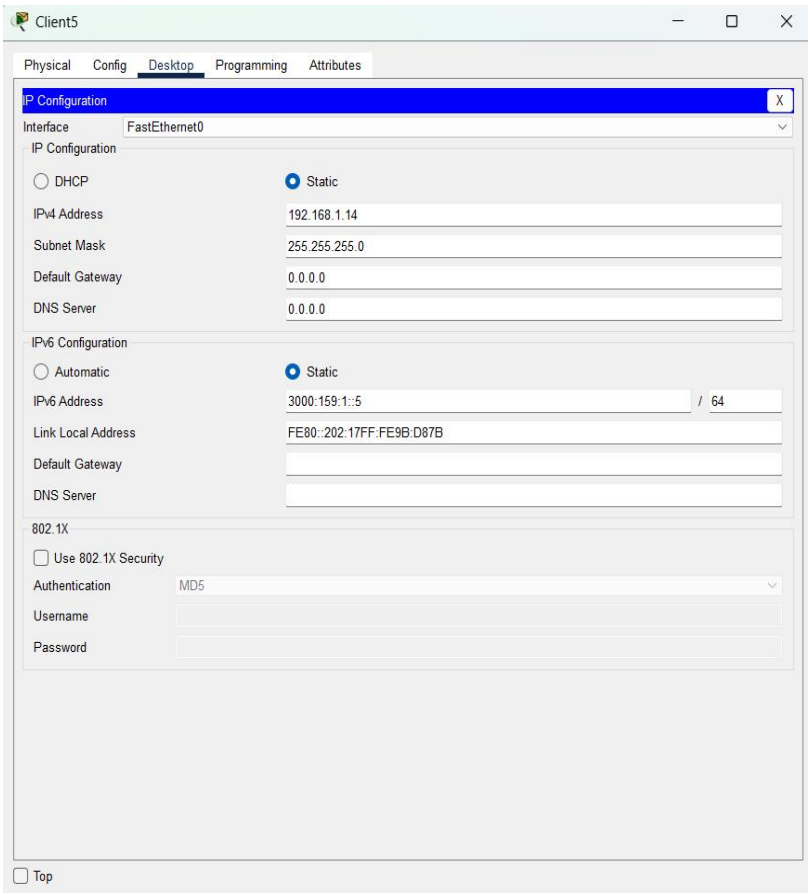
a) Configuration de l’adresses IPv6 des PC

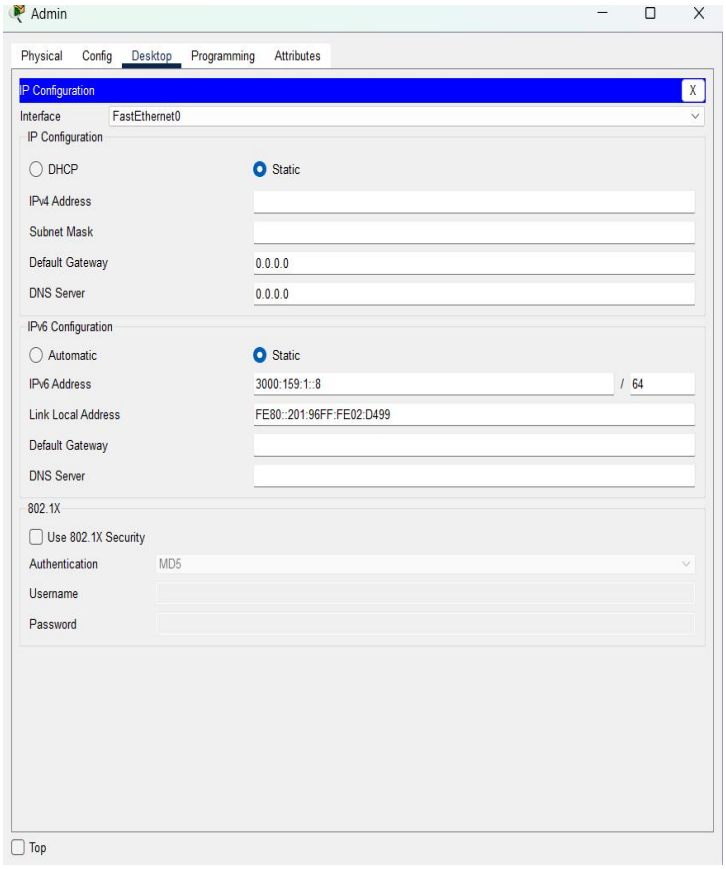




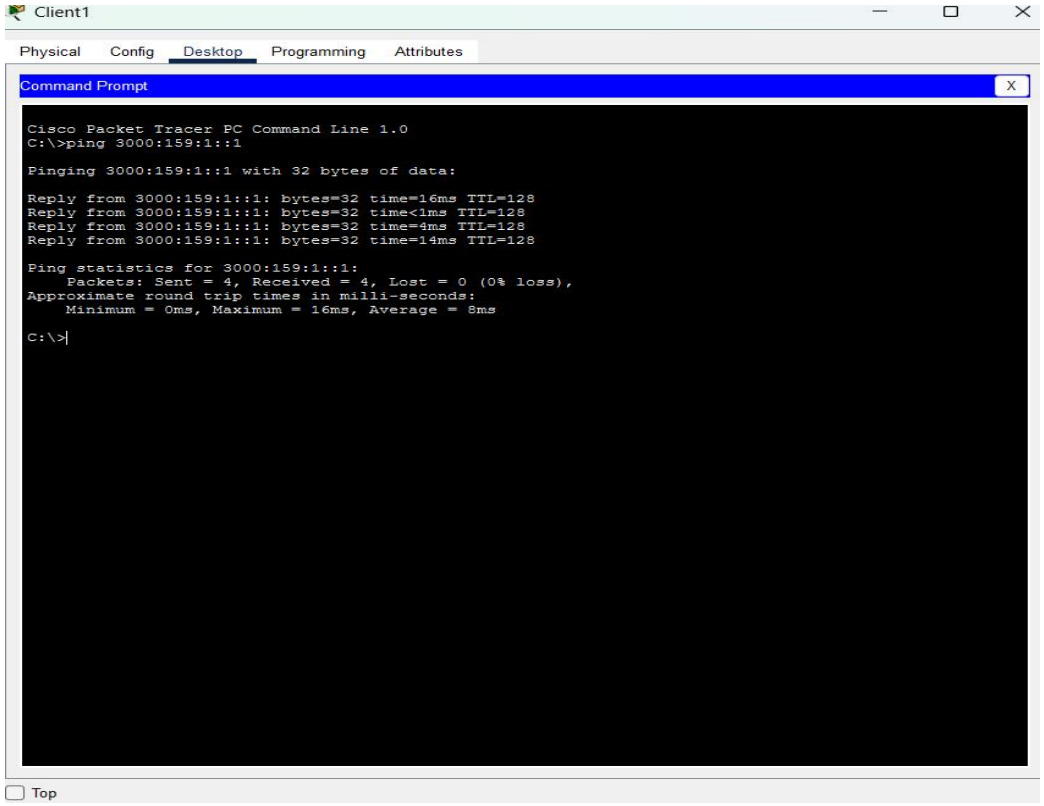

Client3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::2D0:FFFF:FE9D:6A13
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


client4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::4
/ 64
Link Local Address
FE80::201:42FF:FEA5:750
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

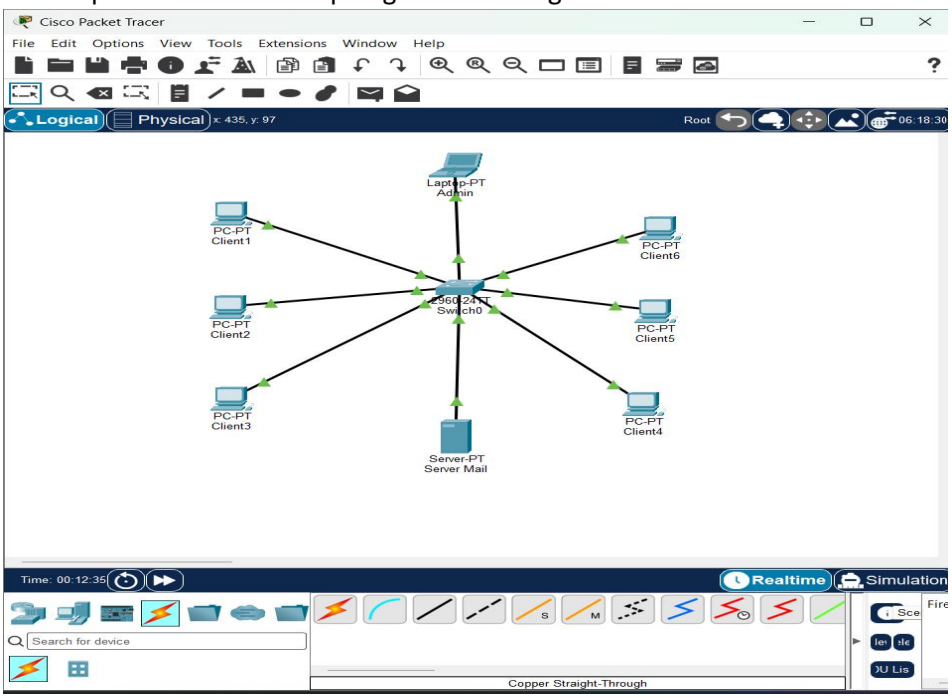

Client5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::5
/ 64
Link Local Address
FE80::202:17FF:FE9B:D87B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Admin
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::8
/ 64
Link Local Address
FE80::201:96FF:FE02:D499
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

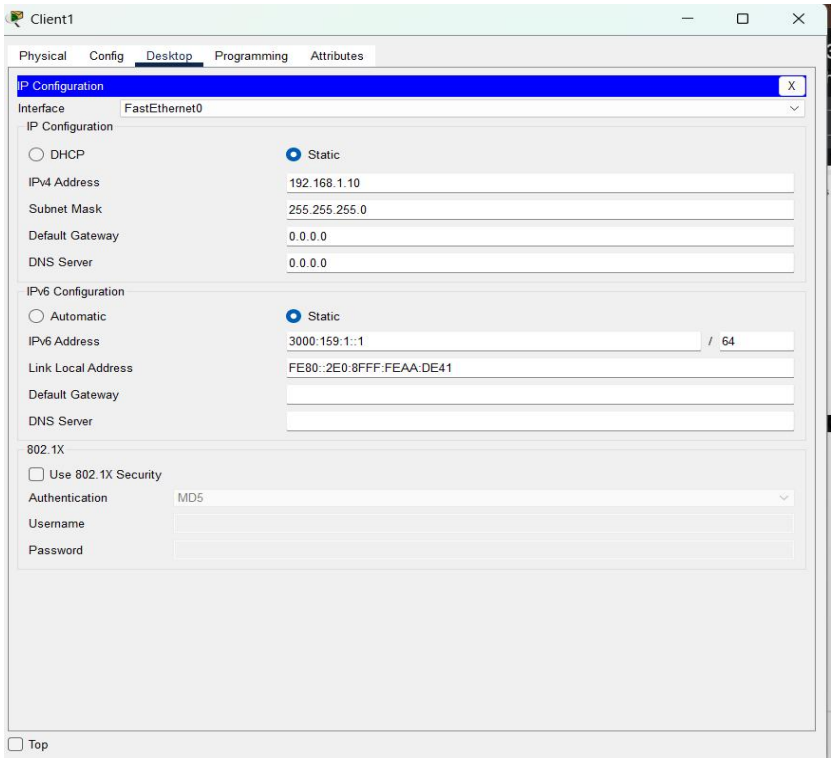
b) Teste de la connexion

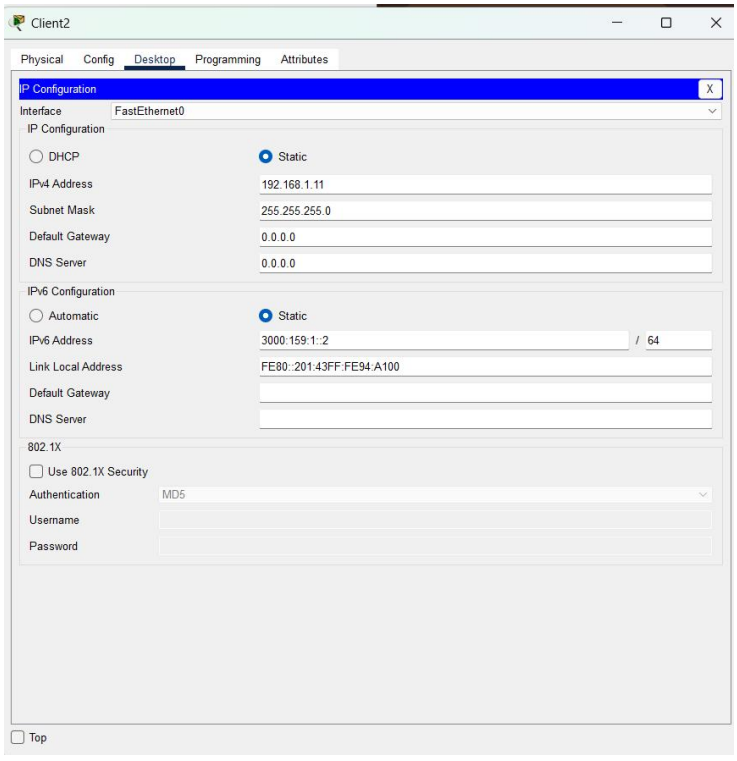


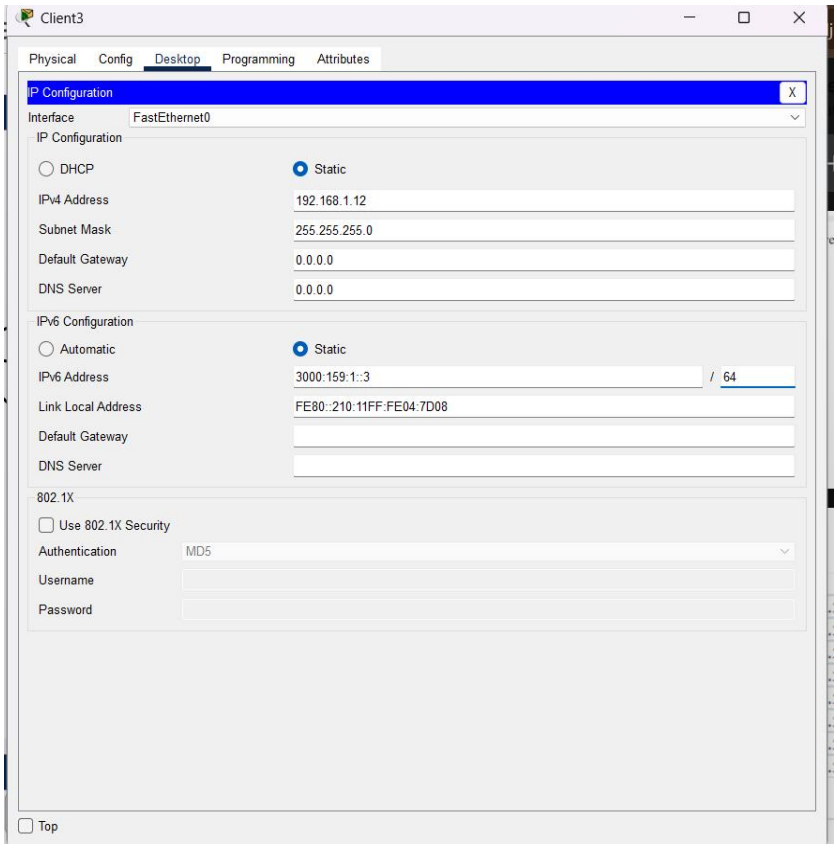
5- La reproduction d'une topologie en le configurant IPv6

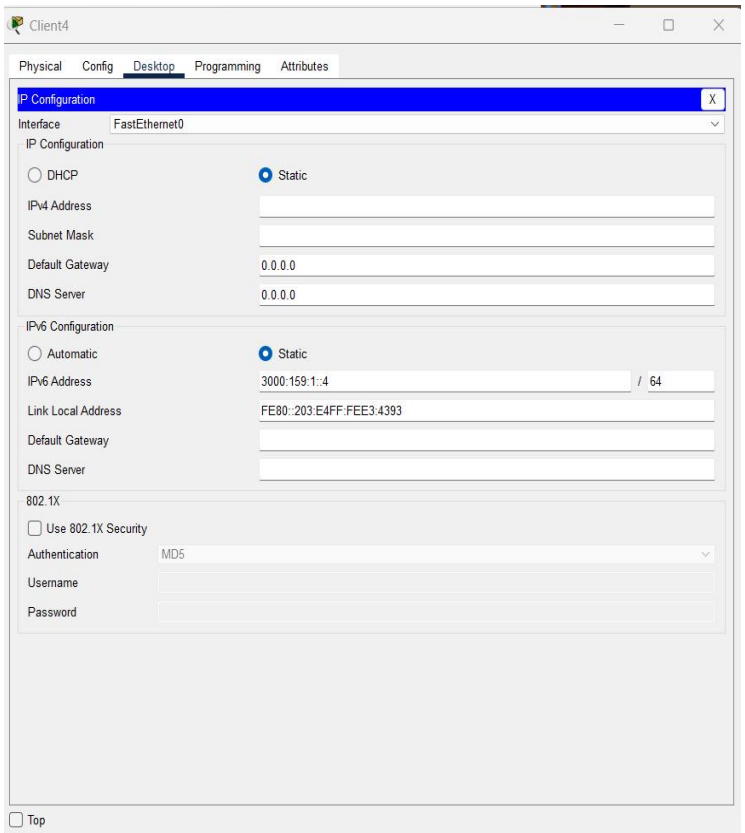


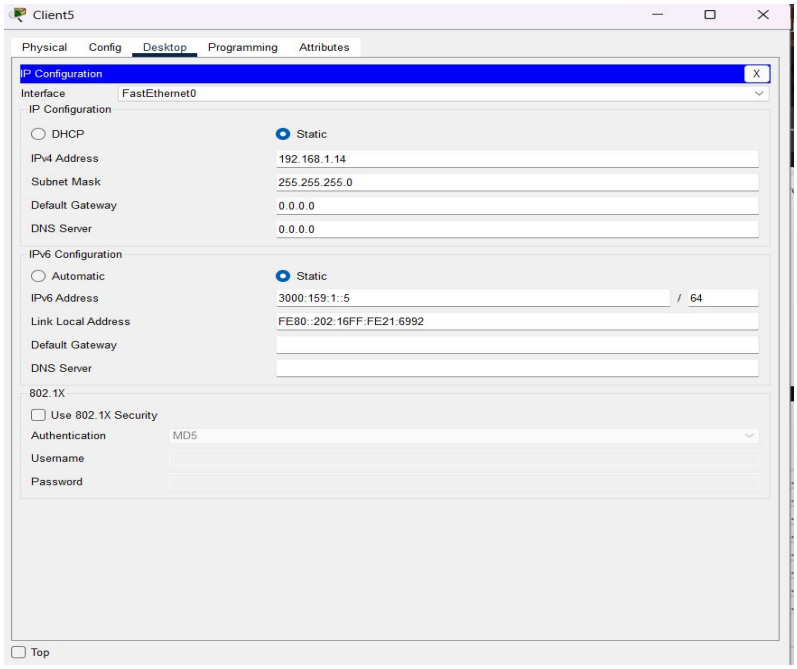
a) configuration de l'adresses IPv6 des PC

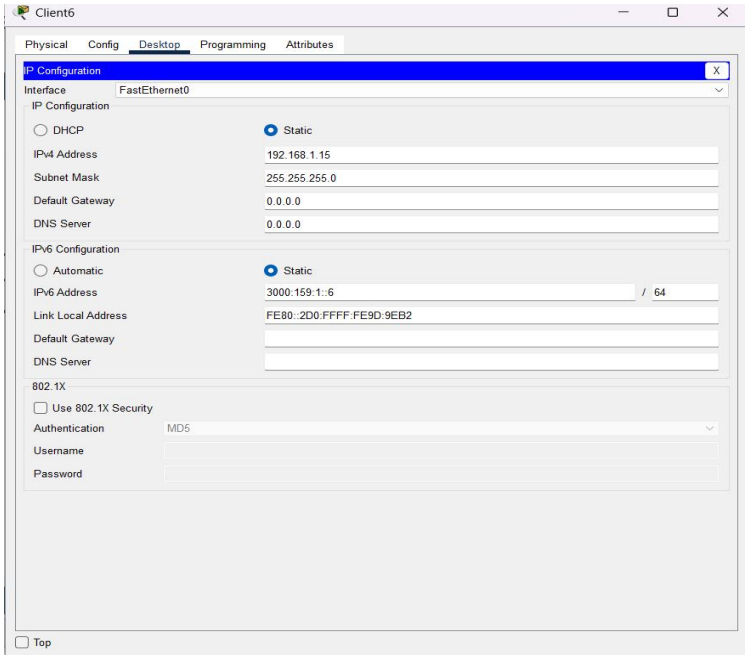

Client1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::1
/ 64
Link Local Address
FE80::2E0:8FFF:FEAA:DE41
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

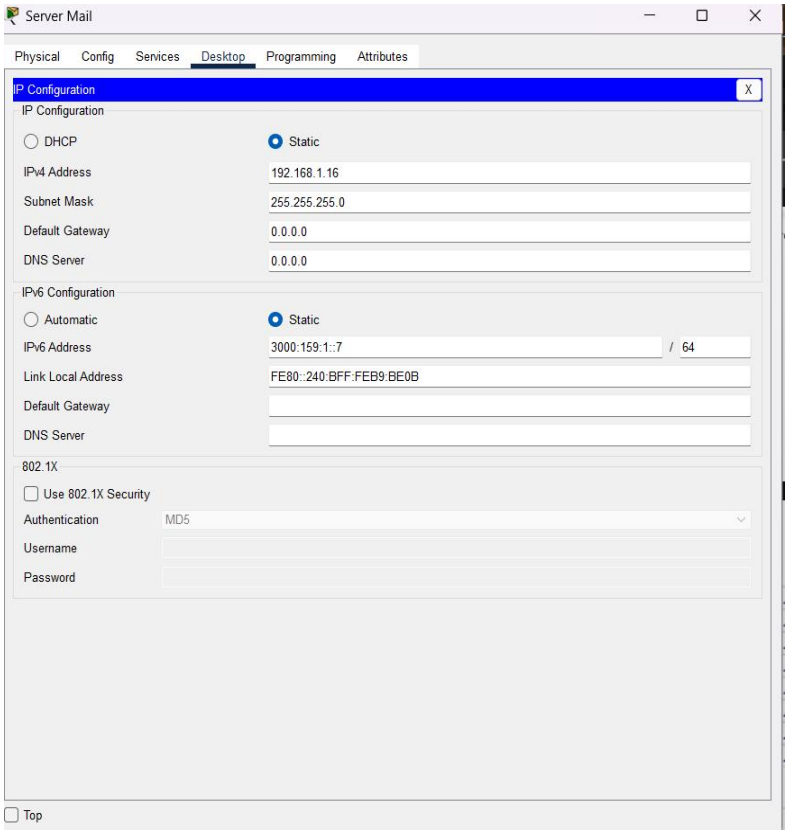

Client2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::2
/ 64
Link Local Address
FE80::201:43FF:FE94:A100
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

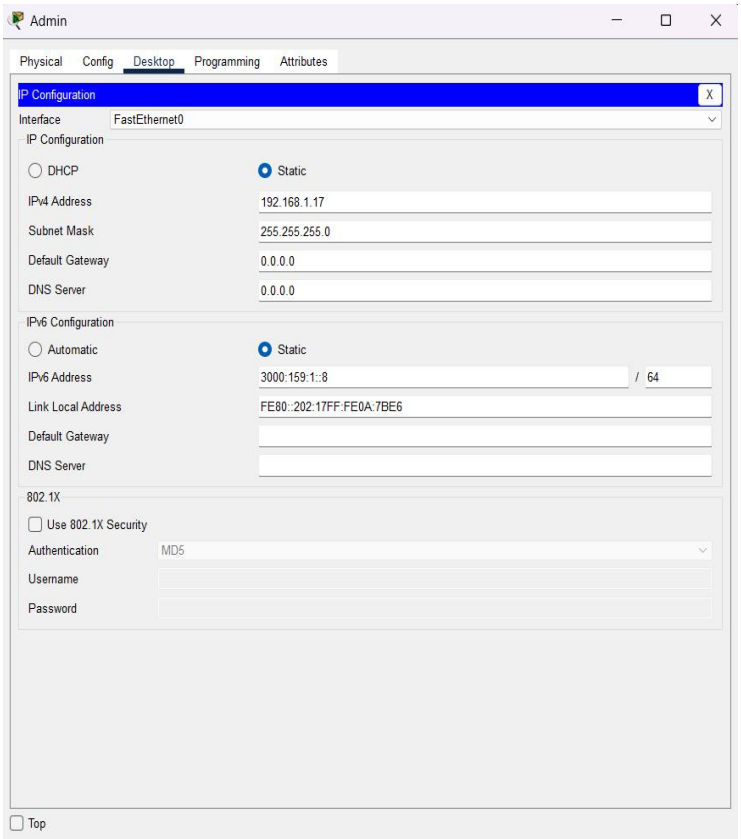

Client3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::210:11FF:FE04:7D08
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Client4
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::4
/ 64
Link Local Address
FE80::203:E4FF:FEE3:4393
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

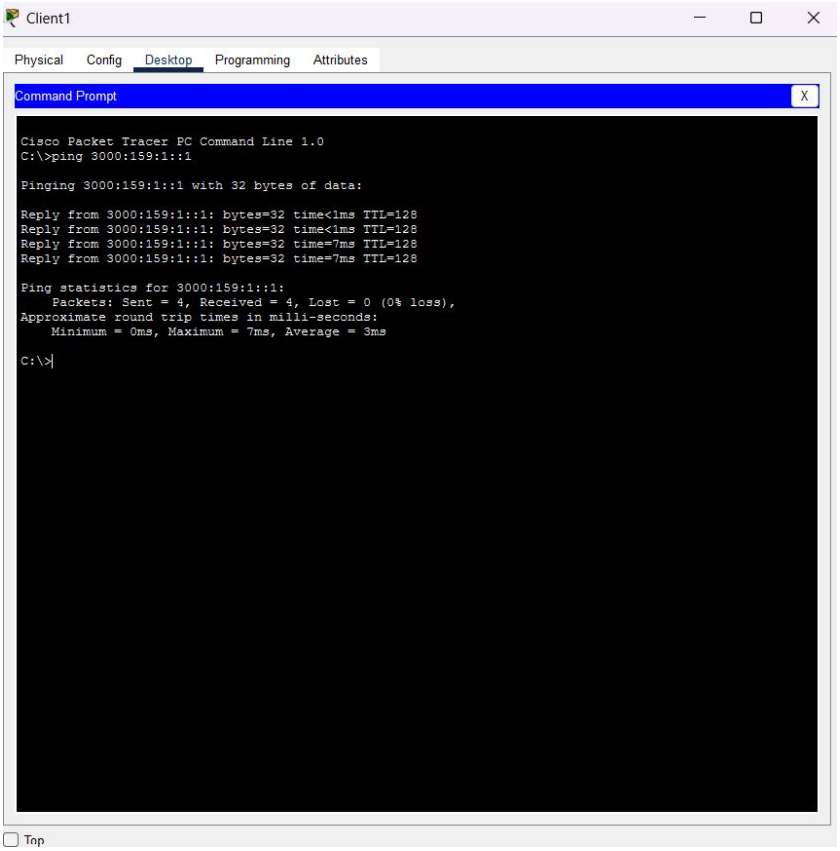

Client5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.14
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::5
/ 64
Link Local Address
FE80::202:16FF:FE21:6992
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Client6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.15
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::6
/ 64
Link Local Address
FE80::2D0:FFFF:FE9D:9EB2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

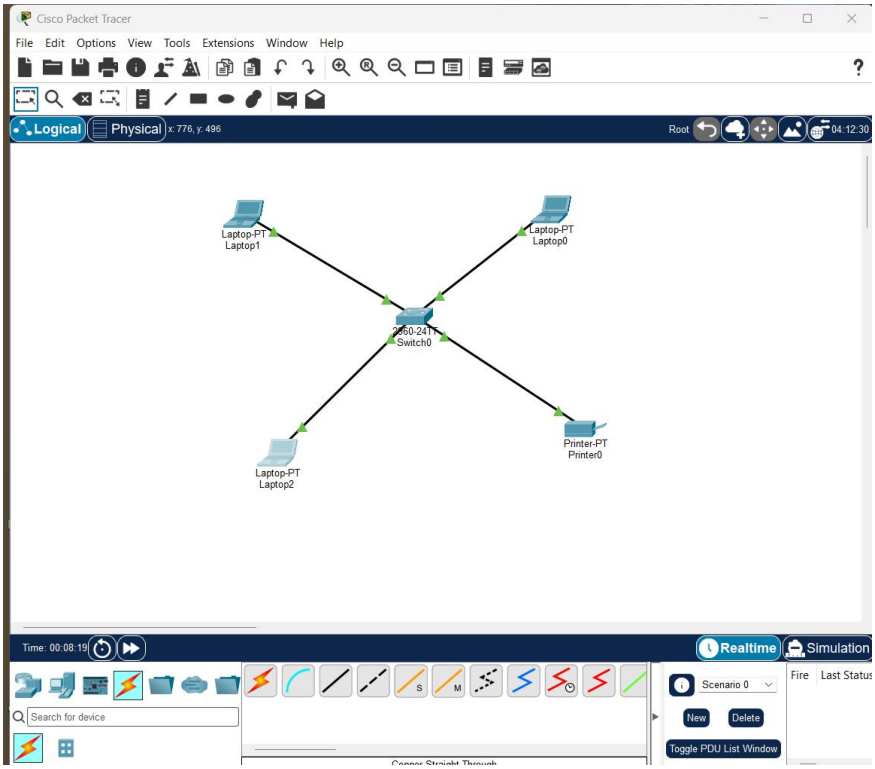

Server Mail
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.16
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::7
/ 64
Link Local Address
FE80::240:BFF:FEB9:BE0B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Admin
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.17
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::8
/ 64
Link Local Address
FE80::202:17FF:FE0A:7BE6
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

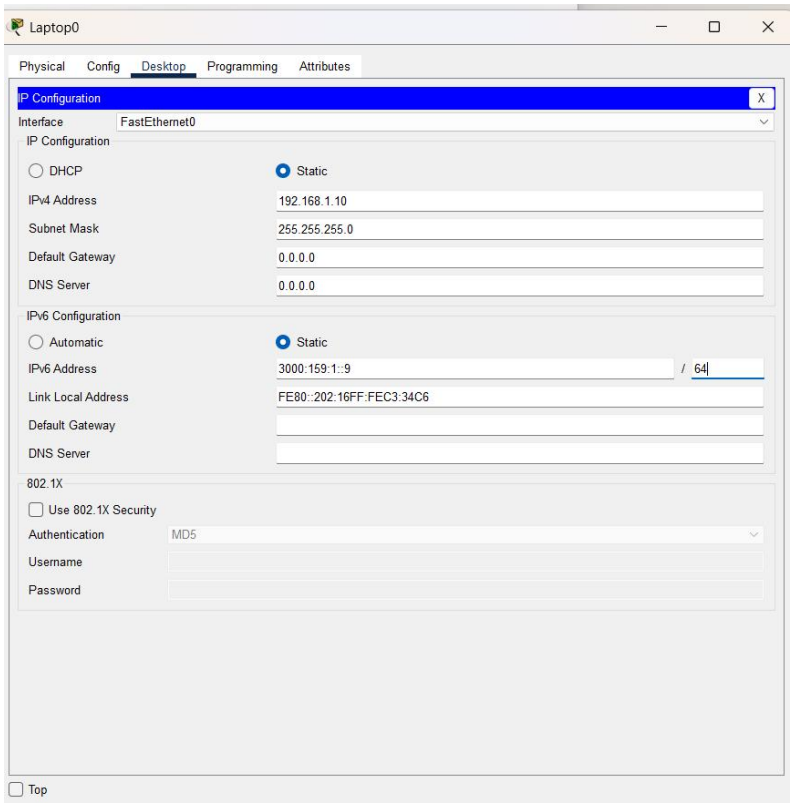
b) teste de connexion

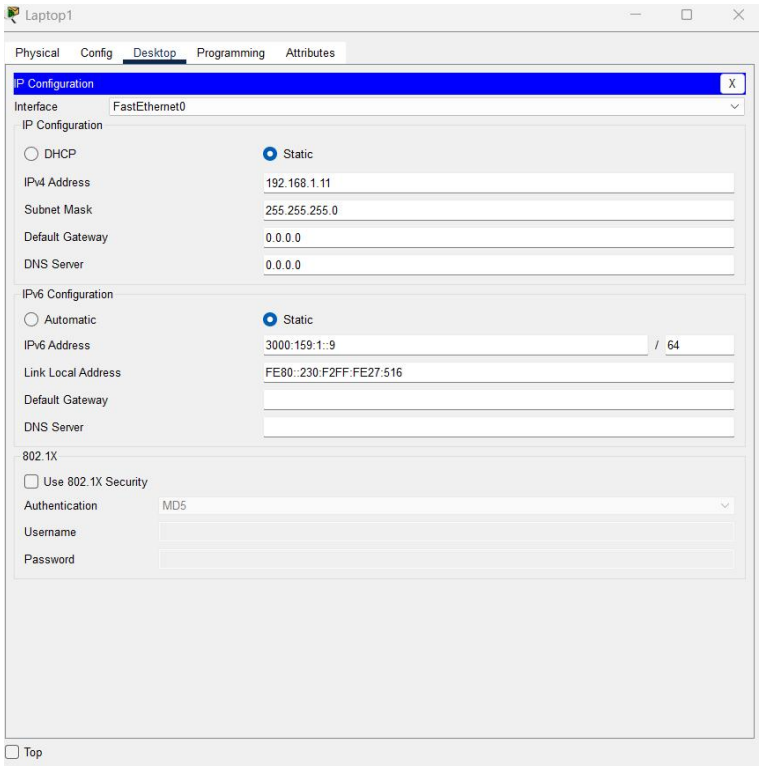


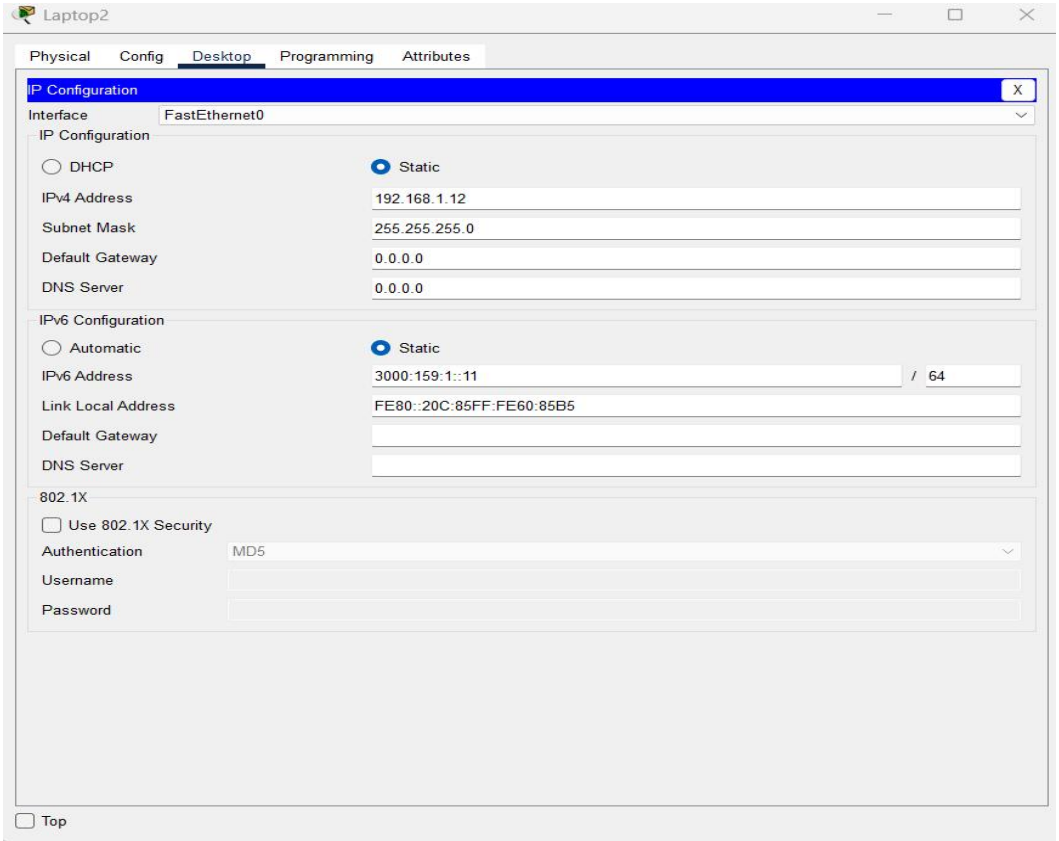
6- parmi les topologies, choisissez-en une, configurant les adresses IPv6 et tester la connectivité



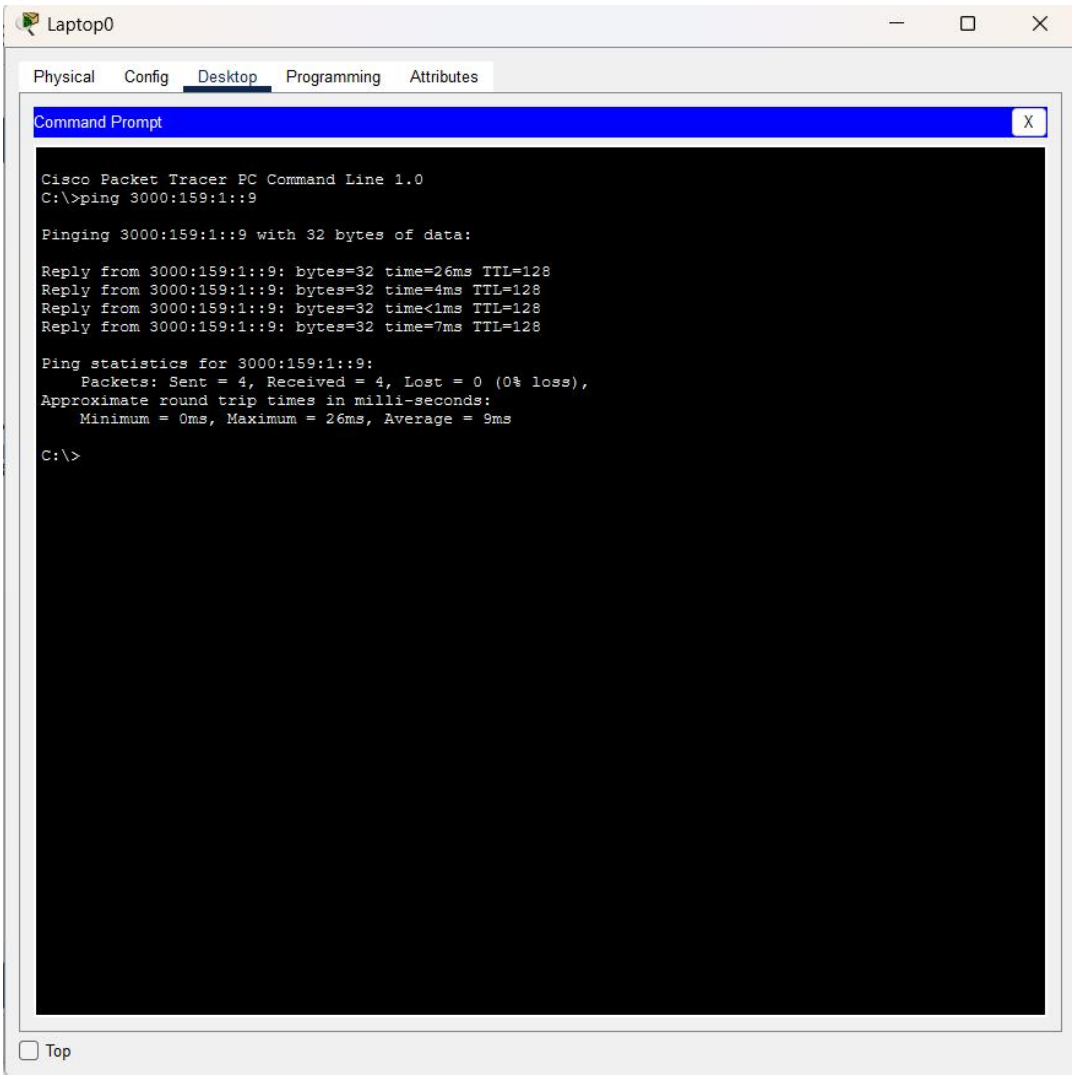
a) Configuration de l’adresses IPv6 des laptop







b) Teste de connexion



Conclusion

on peut conclure pour dire que nous avons appris beaucoup de chose en faisant ce TD, par exemple savoir comment configurer un adresses IPv4 puis IPv6 et comment on peut teste la connexion.